Prior Selection for VAR Forecasts

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Abstract

Selecting appropriate shrinkage priors is crucial for forecasting with parameter-rich VAR models. However, the method proposed by Giannone et al. (2015) faces two key limitations: (i) sensitivity to the choice of hyperprior, and (ii) excessive weight on in-sample fit in the likelihood. To address these issues, this paper employs Relative Entropy Prior Sensitivity (REPS), which is particularly well-suited to our analysis due to (i) its applicability to models of any size, and (ii) its ability to assess the role of the likelihood in hyperparameter estimation. I find that REPS-based hyperparameters alleviate the limitations of Giannone et al. (2015) and improve point forecast accuracy particularly: (i) in small- and medium-sized VARs; (ii) at longer forecast horizons; (iii) in small-sized VARs during COVID. These improvements are statistically significant. I also find that these patterns are attributed to the shape of the marginal likelihood. The findings underscore the importance of prior selection in VAR forecasting across different model sizes and periods. Moreover, the paper provides a systematic framework for enhancing prior selection in Bayesian VAR forecasts.

1 Introduction

Forecasting macroeconomic variables is important for policymakers, because policy decisions depend not only on current economic conditions but also on expected future developments. Since Sims (1980), vector autoregressions (VARs) have been widely used for this purpose because they provide a flexible way to summarize the dynamic relations among macroeconomic variables.

A well-known difficulty of VAR models is that they are rich in parameters. This problem becomes more severe as the number of variables increases. Bayesian VARs address this issue by introducing shrinkage priors, which reduce estimation uncertainty by shrinking VAR coefficients toward more parsimonious specifications. This is especially important for medium- and large-sized VARs, where the number of coefficients can be large relative to the sample size. Prior shrinkage therefore plays a central role in allowing VARs to use larger information sets without losing forecast accuracy.

However, the forecasting performance of Bayesian VARs depends not only on the choice of a prior family, but also on the choice of the hyperparameters governing the strength of shrinkage. In the Minnesota prior, the overall tightness parameter, λ, determines how strongly the VAR coefficients are shrunk toward the prior mean. A larger λ allows the data to speak more, while a smaller λ imposes tighter shrinkage. Therefore, the value of λ affects the trade-off between model flexibility and estimation uncertainty. If λ is too large, the model may overfit the estimation

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sample. If λ is too small, the model may be too tightly restricted. This makes hyperparameter selection a key part of Bayesian VAR forecasting.

Giannone, Lenza, and Primiceri (2015; hereafter GLP) propose a widely used approach to select these hyperparameters. Instead of fixing hyperparameters subjectively, they treat them as unknown parameters and place prior distributions over them. The hyperparameters are then estimated using the marginal likelihood. This approach has become a standard method because it is data-based, easy to implement under conjugate priors, and avoids choosing shrinkage parameters directly.

At the same time, the GLP approach raises two concerns for forecasting. First, the estimated hyperparameters still depend on the choice of the hyperprior. Different hyperpriors can lead to different posterior distributions of the hyperparameters. Second, the marginal likelihood is not the same as an out-of-sample forecasting objective. It is computed using the estimation sample and can put substantial weight on hyperparameter values that explain the in-sample data well. Therefore, even if a hyperparameter is preferred by the marginal likelihood, it may not be the best choice for out-of-sample forecasts. This concern is particularly relevant when the forecasting objective is multi-step-ahead prediction or prediction during unusual periods.

To address these concerns, this paper proposes using a REPS-based prior selection approach for Bayesian VAR forecasting. I focus on the Minnesota prior and its overall tightness hyperparameter, λ. Relative Entropy Prior Sensitivity (REPS), developed by Ho (2023), is a global prior sensitivity method that uses relative entropy to measure the distance between a baseline prior and an alternative prior. In this paper, I apply REPS to construct an entropy-bounded alternative hyperprior and obtain the corresponding REPS-based λ. I then compare forecasts generated using this REPS-based λ with forecasts generated using posterior draws of λ from GLP across model sizes, forecast horizons, and time periods.

REPS is well-suited to this analysis for three reasons. First, because it measures the distance between the original prior and an alternative prior using relative entropy, it provides a disciplined way to move away from the baseline prior without choosing an arbitrary alternative. Second, the method is computationally feasible for small, medium, and large VARs, which makes it suitable for comparing models of different sizes. Third, the REPS solution depends explicitly on the likelihood associated with each hyperparameter value. This feature is central to my analysis because it allows me to examine how the marginal likelihood used by GLP shapes the estimated hyperparameter.

Empirically, I use the FRED-QD dataset and forecast four macroeconomic variables: real GDP growth, GDP deflator inflation, the unemployment rate, and the federal funds rate. I consider small, medium, and large VARs, one- to eight-quarter-ahead forecasts, and several forecast periods, including samples with and without the financial crisis and COVID period. Point forecasts are evaluated using RMSFE, and density forecasts are evaluated using log predictive scores.

I find that the REPS-based hyperparameter improves point forecast accuracy relative to GLP particularly in small- and medium-sized VARs. In large VARs, the improvements are concentrated in persistent variables, particularly the unemployment rate and the federal funds rate. The gains are most pronounced at longer forecast horizons and in small VARs during COVID.

Tests of equal forecast accuracy show that many of these improvements are statistically significant. For density forecasts, the results are more mixed. Since the REPS-based λ is generally smaller than the GLP posterior mean, it imposes stronger shrinkage and produces more concentrated predictive densities. This can improve log scores when point forecasts are accurate, but it can reduce log scores when the predictive density becomes too tight around an inaccurate forecast.

To understand why REPS improves point forecasts, I decompose the marginal likelihood used in GLP into an in-sample fit component and a forecast accuracy component. I find that the REPS-based λ reduces the in-sample fit component but improves the forecast accuracy component. This suggests that REPS improves forecasts by shifting the balance embedded in the marginal likelihood away from in-sample fit and toward forecasting performance.

I also show that the size of the forecast gains depends on the shape of the marginal likelihood. In small and medium VARs, the marginal likelihood is more dispersed, so REPS can move the hyperparameter more and generate larger changes in forecast performance. In large VARs, the marginal likelihood is more concentrated around low values of λ, so REPS induces smaller changes and forecast gains are concentrated in persistent variables such as the unemployment rate and the federal funds rate. During COVID, the likelihood is high and relatively flat over a range of small λ values. As a result, REPS can generate a large hyperparameter distortion, but the resulting improvement in forecast accuracy is limited.

This paper is related to three strands of literature. First, it contributes to the literature on Bayesian VAR forecasting with shrinkage priors. Banbura et al. (2010) show that large Bayesian VARs can forecast well when shrinkage is adjusted to model size. Koop (2013) compares medium and large Bayesian VARs with alternative priors, including the stochastic search variable selection prior. More recent work studies global-local shrinkage priors and adaptive shrinkage methods. This paper differs from these studies by keeping the prior family fixed and focusing on how to select the hyperparameter of the standard Minnesota prior.

Second, this paper contributes to the literature on hyperparameter selection in Bayesian VARs. GLP provide a hierarchical method that selects hyperparameters through the marginal likelihood. This paper builds on their framework but asks whether the hyperparameter preferred by the marginal likelihood is always appropriate for forecasting. The results suggest that alternative hyperparameters can improve out-of-sample point forecasts, especially when the marginal likelihood places strong weight on in-sample fit.

Third, this paper contributes to the literature on prior sensitivity in Bayesian forecasting. Feldkircher (2012), Chan et al. (2019), and Jacobi et al. (2025) study how Bayesian forecasts and other posterior quantities respond to prior assumptions. Ho (2023) develops a global robust Bayesian analysis method that is feasible in large models. This paper applies a global prior sensitivity tool to Bayesian VAR forecasting and uses it to construct forecast-relevant hyperparameter distortions. The paper has three main contributions. First, it shows that entropy-based prior distortions can be used to improve hyperparameter selection in Bayesian VAR forecasting. Second, it provides evidence that the marginal likelihood can select hyperparameters that fit the estimation sample well but are not necessarily best for out-of-sample forecasts. Third, it documents that the effectiveness of hyperparameter distortion depends systematically on model size, forecast horizon, and time period through the shape of the marginal likelihood.

The remainder of the paper is organized as follows. Section 2 introduces the VAR model and the Minnesota prior. Section 3 describes the forecasting methodology and forecast evaluation metrics. Section 4 presents the REPS framework. Section 5 describes the data. Section 6 discusses the empirical results, and Section 7 concludes.

2 Model

2.1 VAR Model

In this paper, I study the following VAR model with p lags:

$$\begin{aligned} y_t &= C + B_1 y_{t-1} + \cdots + B_p y_{t-p} + u_t \\ u_t &\sim \mathcal{N}(0, \Sigma) \end{aligned}$$

where y_t is a $N \times 1$ vector of endogenous variables, u_t is a $N \times 1$ vector of exogenous shocks, and C, B_j for $j = 1, \cdots, p$ and Σ are of dimensions $N \times 1$, $N \times N$, and $N \times N$, respectively, with unknown parameters.

Stacking over T observations, I obtain:

$$\underbrace{\begin{pmatrix} y'_t \\ \vdots \\ \vdots \\ y'_{t+T-1} \end{pmatrix}}_{Y} = \underbrace{\begin{pmatrix} y'_{t-1} & \cdots & y'_{t-p} & 1 \\ \vdots & \vdots & \vdots & \vdots \\ \vdots & \vdots & \vdots & \vdots \\ y'_{t+T-2} & \cdots & y'_{t+T-1-p} & 1 \end{pmatrix}}_{X} \underbrace{\begin{pmatrix} B'_1 \\ \vdots \\ B'_p \\ C' \end{pmatrix}}_{B} + \underbrace{\begin{pmatrix} u'_t \\ \vdots \\ \vdots \\ u'_{t+T-1} \end{pmatrix}}_{U}$$

$$Y = XB + U$$

where Y is a $T \times N$ matrix of endogenous variables, X is a $T \times (Np + 1)$ matrix of regressors, U is a $T \times N$ matrix of shocks, and B is a $(Np + 1) \times N$ matrix with unknown parameters.

I focus on prior distributions for VAR coefficients belonging to the following normal-inverse-Wishart family:

$$\begin{aligned} \Sigma &\sim \mathcal{IW}_N(\underline{S}, \underline{v}) \\ B|\Sigma &\sim \mathcal{MN}_{(Np+1)\times N}(\underline{B}, \underline{V}, \Sigma) \\ \text{vec}(B)|\Sigma &\sim \mathcal{N}_{(Np+1)N\times 1}(\text{vec}(\underline{B}), \Sigma \otimes \underline{V}) \end{aligned}$$

where $\underline{B}$ is a $(Np + 1) \times N$ prior mean matrix, $\underline{V}$ is a $(Np + 1) \times (Np + 1)$ matrix governing column-specific variances, $\underline{S}$ is an $N \times N$ scale matrix, and $\underline{v}$ is the degrees of freedom parameter.

$\mathbf{\Sigma}$: For the prior on Σ, to guarantee the existence of its prior mean, I follow Kadiyala and Karlsson (1997) and set the degrees of freedom for the inverse-Wishart distribution to $\underline{\text{v}} = N+2$. Additionally, I take $\underline{\text{S}}$ to be a diagonal matrix with $\hat{\sigma}^2$ on the diagonal, where $\hat{\sigma}^2$ is estimated as the residual variance from a univariate AR(1) process.

B : For the prior on B, I adopt the *Minnesota prior*, originally proposed by Doan et al. (1984). This prior is motivated by the empirical observation that many macroeconomic time series behave like unit root processes. It imposes shrinkage that helps reduce the complexity of parameter-rich VAR models by shrinking them toward more parsimonious specifications. I discuss this prior in greater detail in Section 2.2, *Shrinkage priors*.

The posterior distribution is then given by:

$$\Sigma|Y \sim \mathcal{IW}_N(\bar{S}, \bar{v})$$
$$B|Y, \Sigma \sim \mathcal{MN}_{(Np+1)\times N}(\bar{B}, \Sigma, \bar{V})$$

where:

$$\bar{B} = \bar{V}(\underline{V}^{-1}\underline{B} + X^\top Y), \quad \bar{V} = (\underline{V}^{-1} + X^\top X)^{-1}$$
$$\bar{S} = \underline{S} + Y^\top Y + \underline{B}^\top \underline{V}^{-1}\underline{B} - \bar{B}^\top \bar{V}^{-1}\bar{B}, \quad \bar{v} = \underline{v} + T$$

This paper adheres to the definitions of small, medium, and large VARs provided by GLP. This is different from the classification in Banbura et al. (2010), which could be explored in future research.

1. *SMALL*

   This is a monetary VAR with three key variables —GDP, the GDP deflator, and the federal funds rate— plus the unemployment rate.

2. *MEDIUM*

   This medium-scale VAR includes the eight variables used in the estimation of the DSGE model by Smets and Wouters (2007) for the U.S. economy: real consumption, real investment, hours worked, wages, and the unemployment rate.

3. *LARGE*

   This large-scale VAR includes 23 variables that encompass the previous two specifications and additionally incorporate several important labor market, financial, and monetary variables.

2.2 Shrinkage Priors

VARs tend to be rich in parameters, and this problem intensifies with the size of the model. To mitigate this problem, researchers often introduce shrinkage priors that "shrink" the richly parameterized unrestricted VAR model toward a more parsimonious specification. Here, I introduce the most widely used shrinkage prior for the VAR coefficients—the Minnesota prior.

The *Minnesota prior* is motivated by the empirical observation that many macroeconomic time series behave approximately as unit root processes. Accordingly, the prior mean on the VAR coefficients is constructed to reflect this belief by centering the prior mean for the first own-lag of each variable around one, while shrinking all other coefficients toward zero.

$$\mathbb{E}(\mathrm{B}_\ell^{ij}) = \begin{cases} 1 & \text{if } i = j \text{ and } \ell = 1 \\ 0 & \text{otherwise} \end{cases}$$

where ℓ is the lag order, and $i, j = 1, \cdots, N$. This implies the following prior mean matrix for the system of VAR equations:

$$\underline{\mathbf{B}} = \left(I_N; 0_{N\times N(p-1)}; 0_{1\times N}\right)$$

The matrix $\underline{V}$ determines the prior covariance matrix of B and is assumed to be diagonal, reflecting the lack of strong prior beliefs about correlations between parameters.

$$\Sigma \otimes \underline{V} = \begin{pmatrix} \Sigma_{11}\underline{V} & \dots & \Sigma_{1N}\underline{V} \\ \vdots & \vdots & \vdots \\ \Sigma_{N1}\underline{V} & \dots & \Sigma_{NN}\underline{V} \end{pmatrix}$$

$$= \begin{pmatrix} \Sigma_{11}\begin{pmatrix} \underline{V}_{11} & 0 & 0 \\ \vdots & \ddots & \vdots \\ 0 & 0 & \underline{V}_{(Np+1)(Np+1)} \end{pmatrix} & \dots & \Sigma_{1N}\begin{pmatrix} \underline{V}_{11} & 0 & 0 \\ \vdots & \ddots & \vdots \\ 0 & 0 & \underline{V}_{(Np+1)(Np+1)} \end{pmatrix} \\ \vdots & \vdots & \vdots \\ \Sigma_{N1}\begin{pmatrix} \underline{V}_{11} & 0 & 0 \\ \vdots & \ddots & \vdots \\ 0 & 0 & \underline{V}_{(Np+1)(Np+1)} \end{pmatrix} & \dots & \Sigma_{NN}\begin{pmatrix} \underline{V}_{11} & 0 & 0 \\ \vdots & \ddots & \vdots \\ 0 & 0 & \underline{V}_{(Np+1)(Np+1)} \end{pmatrix} \end{pmatrix}$$

The diagonal elements of $\underline{V}$ govern the prior variances of the VAR coefficients. In the Minnesota prior, the prior reflects beliefs about the relative importance of lags and cross-variable relationships. Specifically, it assigns smaller variances to coefficients on more distant lags, reflecting the belief that higher-order lags are less important. In addition, it rescales the prior variances of coefficients based on the relative variances of the explanatory and dependent variables.

Formally, the diagonal elements of $\underline{V}$ are given by:

$$\underline{\mathbf{V}}_{ii} = \frac{\lambda}{\ell^2 \hat{\sigma}_i^2} \text{ where } i = (\ell - 1)N + n + 1,\ \ \ell = 1, \cdots, p,\ \ n = 1, \cdots, N$$

where ℓ is the lag order, $\hat{\sigma}_i^2$ is the estimated residual variance from an AR(1) model for variable i, and λ is a hyperparameter governing overall tightness.

Combining this with the error variance Σ_{jj} from the jth equation, the implied prior variance for the coefficient on the pth lag of variable i in jth equation is:

$$\mathrm{Var}(\mathrm{B}_p^{ij}) = \frac{\lambda\, \hat{\sigma}_j^2}{p^2\, \hat{\sigma}_i^2}$$

The key hyperparameter is λ, which controls the overall tightness of the prior by scaling the variances of the coefficient distributions. As such, the choice of λ is critical.

Given the central role of λ in shrinking the VAR models, the main analysis focuses exclusively on cases where the Minnesota prior is used. The analysis incorporating the dummy-initial-observation priors alongside the Minnesota prior is presented in the appendix.

2.3 Prior Selection: Hyperparameter Estimation by GLP

Prior to GLP, economists typically adopted values of λ from previous studies, even when using different datasets or model sizes. Instead, GLP treat λ as an unknown parameter and estimate it by placing a prior distribution over it. Following their specification, I impose a Gamma hyperprior with a mode of 0.2 and a standard deviation of 0.4

When estimating hyperparameters using the Metropolis-Hastings algorithm, GLP use the following marginal distribution as the likelihood for the hyperparameters. This is referred to as the marginal likelihood of the data as a function of the hyperparameters, since it is obtained by integrating out the VAR coefficients.

$$p(y|\lambda) = \int p(y|\theta,\lambda)\ p(\theta|\lambda)\ d\theta$$
$$\propto \underbrace{\left|\left(V_\epsilon^{\text{posterior}}\right)^{-1} V_\epsilon^{\text{prior}}\right|^{-\frac{T-p+d}{2}}}_{\text{In-Sample Fit}} \quad \cdot \quad \underbrace{\prod_{t=p+1}^{T} \left|V_{t|t-1}\right|^{-\frac{1}{2}}}_{\text{Forecast Accuracy}}$$

where θ denotes the VAR coefficients, λ represents the tightness hyperparameter, $V_\epsilon^{\text{posterior}}$ and $V_\epsilon^{\text{prior}}$ are the posterior and prior means of the residual variance, and $V_{t|t-1}$ is the variance of the one-step-ahead forecast of y, averaged prior draws of Σ. They argue that this method effectively selects hyperparameters by balancing the trade-off between in-sample fit and forecast accuracy. Moreover, it admits a closed-form expression under conjugate priors, allowing for easy numerical implementation.

However, GLP's method raises two concerns. First, it remains dependent on the choice of the hyperprior—an alternative hyperprior may lead to different hyperparameter values. Second, and more importantly, it is questionable whether this choice of hyperparameters is suitable for forecasting practice. This is because the likelihood used to estimate hyperparameters is claimed to balance forecast accuracy with in-sample fit, but this tradeoff may be problematic. Specifically, the likelihood places excessive weight on in-sample fit, which can lead the model to overfit the estimation sample, resulting in poor out-of-sample performance. Therefore, alternative hyperparameters may yield better out-of-sample forecasts, albeit at the cost of in-sample fit. To explore this possibility, this paper proposes using the Relative Entropy Prior Sensitivity tool introduced by Ho (2023), which is discussed in detail in Sections 4 and 6.

I generated 10^5 hyperparameter prior and posterior draws for each estimation sample. Posterior draws were obtained using the Metropolis-Hastings algorithm, with 2×10^4 burn-in draws. This was followed by 10^6 additional draws, of which one in ten was retained.

For each draw of the posterior hyperparameters, I then estimated the VAR coefficients using closed-form expressions for their posterior distribution. Consequently, I generated 10^5 estimates of the VAR coefficients for each estimation sample. Alternatively, one can use the posterior mean of the hyperparameter draws to estimate the VAR coefficients. While this paper focuses on the former approach, results using the latter method are reported in the appendix.

3 Forecasts

This section introduces the forecasting method and the metrics used to evaluate forecast performance. Based on the iterated forecasting approach, I generate one- to eight-quarter-ahead forecasts for real GDP growth, GDP-deflator inflation, the unemployment rate, and the federal funds rate—both point forecasts and density forecasts—using data from the past 20 years. Point forecasts and density forecasts are then evaluated using the root mean squared forecast error (RMSFE) and the log predictive score, respectively.

3.1 Forecasting Method

I forecast four macroeconomic indicators: real GDP growth, GDP deflator inflation, the unemployment rate, and the federal funds rate, using small ($n = 4$), medium ($n = 8$), and large

($n = 23$) VAR models with $p = 4$ lags. Specifically, I employ an iterated forecasting approach with a rolling window to generate a series of one- to eight-step-ahead forecasts for each variable.

Let $\mathbf{y}_{1:T}$ denote the estimation sample. The iterated forecasting method first constructs the companion form of the VAR model and then generates h-step-head forecasts iteratively as follows:

VAR:

$$y_t = C + B_1 y_{t-1} + \cdots + B_p y_{t-p} + u_t, \text{ for } t = p+1, \cdots, T$$

Companion form:

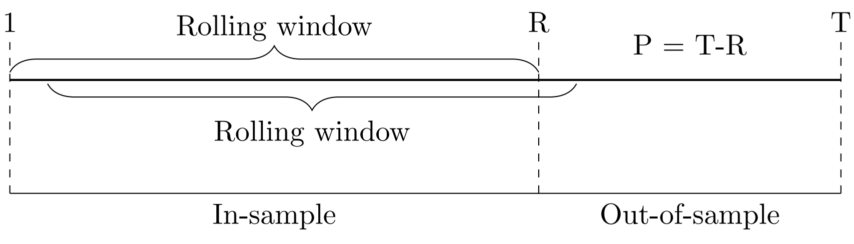
$$\underbrace{\begin{bmatrix} 1 \\ y_t \\ \vdots \\ y_{t-p} \end{bmatrix}}_{X_{t+1}} = \begin{bmatrix} 1 & 0 & \dots 0 & \\ C & B_1 & \dots & B_p \\ 0 & I_N & \dots & 0 \\ \vdots & \vdots & \vdots & \vdots \\ 0 & 0 & 0 & 0 \end{bmatrix} \begin{bmatrix} 1 \\ y_{t-1} \\ \vdots \\ y_{t-p-1} \end{bmatrix} + \begin{bmatrix} 0 \\ u_t \\ \vdots \\ 0 \end{bmatrix}$$

$$X_{t+1} = (B^{+})X_t + \tilde{u}_t$$

Multi-step (h-step-ahead) forecast:

$$\hat{X}_{t+h} = (B^{+})\hat{X}_t + \tilde{u}_{t+h} \text{ for } h = 1, \cdots, 8$$

The second row of X_{t+1} corresponds to a draw from the posterior predictive density of y_{T+h}. Given the 100,000 estimates of the VAR coefficients from the sample $\mathbf{y}_{1:T}$ in Section 2, I generate 100,000 draws from the predictive density of y_{T+h}. I then iterate this procedure by updating the estimation sample by shifting the sample forward one quarter to $\mathbf{y}_{2:T+1}$, re-estimating the VAR, and generating forecasts of y_{T+1+h}. This process is repeated one quarter at a time until the end of the sample.



More precisely, starting from a sample ranging from 1968Q2 to 1988Q1, I generate draws from the posterior predictive density of the model for 1988Q2 (one quarter ahead) and 1990Q1 (two years ahead). I then update the estimation sample, one quarter at a time, until the end of the sample, 2023Q4. At each rolling window, I re-estimate the model and produce F_{draws} from the posterior predictive density.

I repeat this procedure for four different periods: Sample 1, Sample 2, the financial crisis, and COVID. This procedure yields F_{draw}, a time series of P density forecasts for each variable and forecast horizon. I use the median of each distribution as the point forecast.

Table 1. Forecasting Periods Outline

Periods	t=1	t=R+1	t=T-7	t=T	P
Sample 1	Apr, 1968	Apr, 1988	Apr, 2006	Jan, 2008	73
Sample 2	Apr, 1968	Apr, 1988	Jan, 2023	Oct, 2024	140
Financial Crisis	Oct, 1987	Oct, 2007	Oct, 2015	Oct, 2017	33
COVID	Jan, 2000	Jan, 2020	Jan, 2022	Oct, 2023	9

(1) Entire Sample: Sample 1 excludes the financial crisis and COVID periods by ending before the financial crisis, whereas Sample 2 includes all available data.

(2) Unusual Periods: I define unusual periods (e.g. financial crisis, COVID) based on the Fed's policy decisions. Specifically, I set the first one-step-ahead forecast for these periods (t=R+1) as the time when the Fed began lowering the federal funds rate. Likewise, I set the last one-step-ahead forecast for these periods (t=T-7) as the time when the Fed started raising the federal funds rate. This definition reflects the role of forecasting in informing policy decisions.

F_{draw} is equal to the number of hyperparameter posterior draws, obtained from the estimation procedure following GLP in Section 2. For each draw, I sample the corresponding VAR coefficients and generate one- to eight-step-ahead forecasts using the iterated method. This results in 100,000 forecasts for each variable i and forecast horizon h across rolling windows. In contrast, Ho's approach provides a single hyperparameter value. Using this fixed value, I generate 100,000 draws of VAR coefficients and construct forecasts based on each draw.

As a result, I obtain the set of h-step ahead forecasts for variable i's at the t th rolling window, represented as $\left\{y_{i,h}^{t,f}\right\}_{f=1}^{100{,}000}$, for both Ho's method and GLP's method. The point forecast, $\hat{y}_{i,h}^{t}$, is defined as the median of this set. Let $y_{i,h}^{t}$ denote the realized value for $t = 1, \cdots, P$. Then, the forecast error is given by $e_{i,h}^{t} = \hat{y}_{i,h}^{t} - y_{i,h}^{t}$ for $t = 1, \cdots, P$.

Algorithm:

1. Generate F_{draw} draws of the hyperparameter λ using the approach of GLP.
2. For each draw of λ, compute the corresponding estimates of the VAR coefficients.
3. Given the VAR coefficient estimates, generate variable i's h-step-ahead forecast using the iterated method.

$\Rightarrow$ Obtain F_{draw} draws from the forecast density, from which point forecasts can be computed.

3.2 Measure of Forecasting Accuracy

3.2.1 Point Forecasts

I measure the accuracy of a point forecast using the root mean squared forecast error (RMSFE) for variable i and forecast horizon h where $i \in \{$real GDP growth, GDP-deflator inflation, unemployment rate, federal funds rate$\}$ and $h = \{1, \cdots, 8\}$. The RMSFE of variable i at horizon h is computed as follows:

$$\text{RMSFE}_{i,h} = \sqrt{\frac{1}{P}\sum_{t=1}^{P}(y_{t+h}^{i} - \hat{y}_{t+h}^{i})^2}$$

where y^i_{t+h} denotes the actual value of variable i at time $t+h$, $\hat{y}^i_{t+h}$ denotes the median of the predictive density of variable i at time $t+h$, P denotes the length of out-of-sample forecast period, and F_{draw} is the number of draws from the predictive density of variable i at horizon h. A lower RMSFE indicates a superior forecast performance.

To compare the point forecast accuracy of two models using different hyperparameter selection methods —Ho's approach (**PH**) versus Giannone et al.'s method (**GLP**)—, I compute the percentage change in RMSFE for variable i at horizon h as follows:

$$\left(1 - \frac{\text{RMSFE}^{\mathbf{PH}}_{i,h}}{\text{RMSFE}^{\mathbf{GLP}}_{i,h}}\right) \times 100$$

A positive value indicates that the **PH** method outperforms **GLP** method in forecast accuracy.

3.2.2 Density Forecasts

I measure the accuracy of density forecasts using the average log predictive score (SCORE) of variable i and horizon h for $i \in$ {real GDP growth, GDP-deflator inflation, unemployment rate, federal funds rate} and $h = \{1, \cdots, 8\}$. The SCORE of variable i at horizon h is computed as:

$$\text{SCORE}_{i,h} = \frac{1}{P}\sum_{t=1}^{P} s_t(y^i_{t+h})$$

where the log predictive score $s_t(y^i_{t+h})$ is computed using a quadratic approximation of the Gaussian predictive distribution, following Adolfson et al. (2007):

$$s_t(y^i_{t+h}) = -0.5\left[\ln(2\pi) + \ln(V^i_{t+h|t}) + \frac{(y^i_{t+h} - \overline{y}^i_{t+h|t})^2}{V^i_{t+h|t}}\right]$$

for $t = 1, \cdots, P$, where, given F_{draw} from the forecast distribution of y^i_{t+h}, $\left\{y^i_{t+h,f}\right\}_{f=1}^{F_{\text{draw}}}$, the posterior mean $\overline{y}^i_{t+h|t}$ and the posterior variance $V^i_{t+h|t}$ are computed from the forecast distribution, and y^i_{t+h} denotes the actual value. Since the log score evaluates the predictive density at the actual value, a higher log score indicates superior forecast performance.

To compare the density forecast accuracy of two models—Ho's method(**PH**) versus Giannone et al.'s method (**GLP**)—I compute the percentage gain in SCORE for variable i at horizon h as follows:

$$(\text{SCORE}^{\mathbf{PH}}_{i,h} - \text{SCORE}^{\mathbf{GLP}}_{i,h}) \times 100$$

A positive value indicates that the **PH** method outperforms the **GLP** method.

3.3 Tests of Equal Forecast Accuracy

Once I have obtained the forecast accuracy metrics, I need a method to assess whether the forecasting performance of the VAR model differs significantly depending on the hyperparameter selection method. To determine the appropriate testing procedure, it is important to note that this paper employs rolling window estimation to generate unconditional forecasts from potentially nested models. Based on this setup, I employ two tests for point forecast evaluation: (i) the Diebold-Mariano (1995) test and (ii) the Clark and West (2007) test. For density forecast evaluation, I adopt the test proposed by Amisano and Giacomini (2007).

3.3.1 Diebold and Mariano (1995) Test

For point forecasts, the use of rolling window estimation justifies the application of the forecast accuracy test proposed by Giacomini and White (2006), which accounts for parameter estimation uncertainty that may persist in this setting. Giacomini and White (2006) show that, for unconditional forecasts, their test statistic is equivalent to that of Diebold and Mariano (1995). Given this equivalence, I proceed with the Diebold-Mariano (1995) test to evaluate forecast accuracy.

Let $d_t = \left(e_{i,h}^{t,\mathbf{PH}}\right)^2 - \left(e_{i,h}^{t,\mathbf{GLP}}\right)^2$ for $t = 1, \ldots, P$, where $e_{i,h}^{t,\cdot}$ denote the forecast error for variable i at horizon h. Let $\bar{d}$ denote the sample mean of d_t, and estimate the variance of d_t, $\mathrm{var}(d)$, using a heteroskedasticity and autocorrelation consistent (HAC) estimator. The Diebold-Mariano test statistic is then given by:

$$\mathrm{DM} = \frac{\bar{d}}{\sqrt{\frac{1}{P}\,\mathrm{var}(d)}} \sim N(0,1).$$

If the null hypothesis of equal forecast accuracy is rejected, it indicates a statistically significant difference in the predictive performance of the two models.

3.3.2 Clark and West (2007) Test

Since the VAR models considered in this paper include the same set of variables and differ only in their hyperparameter values, the Clark and West (2007) test for nested models is also appropriate. I therefore include the Clark and West (2007) test as a complementary evaluation of forecast accuracy under nested models.

When comparing the mean squared forecast errors (MSFE) of nested models, the Clark-West test corrects for the upward bias in the MSFE of the larger model. To implement the test, define:

$$f_{i,h}^t = \left(e_{i,h}^{t,\mathbf{GLP}}\right)^2 - \left[\left(e_{i,h}^{t,\mathbf{PH}}\right)^2 - (\hat{y}_{i,h}^{t,\mathbf{GLP}} - \hat{y}_{i,h}^{t,\mathbf{PH}})^2\right] \text{ for } t = 1, \cdots, P$$

Then, regress $f_{i,h}^t$ on a constant and use the resulting t-statistic for the coefficient on the constant to test whether the MSFEs of two models are equal. Rejection of the null implies that the **PH** model yields a significantly smaller MSFE than the **GLP** model.

3.3.3 Amisano and Giacomini (2007) Test

For density forecasts, this paper uses the Amisano and Giacomini (2007) t-test of equal means, utilizing the log score for each forecast relative to the benchmark.

In the Amisano and Giacomini (2007) test, the test statistic is defined as follows:

$$\mathrm{WLR}_{i,t+h} = \omega(y_{t+h}^{i,\mathrm{st}}) \cdot \left[s_t(y_{t+h}^i)^{\mathbf{PH}} - s_t(y_{t+h}^i)^{\mathbf{GLP}}\right] \text{ for } t = 1, \cdots, P$$

where $y_{t+h}^{i,\mathrm{st}} = (y_{t+h}^i - u_t)/\sigma_t$ is the realized value of variable i at period $t+h$ standardized using the mean and standard deviation of the estimation sample, ω is standard normal density function, $s_t(y_{t+h}^i)$ is a log predictive score of variable i at $t+h$ evaluated at t.

$$t_{i,h} = \frac{\overline{\mathrm{WLR}}_{i,h}}{\hat{\sigma}_{i,h}/\sqrt{P}}$$

where $\overline{\text{WLR}}_{i,h}$ denotes the sample mean of $\overline{\text{WLR}}_{i,t+h}$ and $\hat{\sigma}_{i,h}$ is the variance estimate of $\text{WLR}_{i,t+h}$ for $t = 1, \cdots, P$ following the suggestion by Amisano and Giacomini (2007). If the null hypothesis of equal forecast accuracy is rejected, it indicates a statistically significant difference in the predictive performance of the two models.

4 Relative Entropy Prior Sensitivity (REPS)

To explore how VAR forecasts change across different hyperparameters and to identify alternative hyperparameter values that improve forecast accuracy at ths cost of in-sample fit, I use Relative Entropy Prior Sensitivity (REPS) developed by Ho (2023). This approach is particularly well-suited to our analysis for three key reasons: (i) its function as a tool for analyzing the sensitivity of posterior estimates to prior assumptions, (ii) its applicability to models of any size, and (iii) its ability to analyze the role of the likelihood in estimating hyperparameters.

REPS is a global prior sensitivity analysis tool that is computationally feasible for both small and large models, using relative entropy to measure the distance between distributions. This allows it to overcome the computational limitations of existing prior sensitivity methods. Accordingly, REPS enables us to analyze how sensitive posterior estimates—and resulting VAR forecasts—are to changes in prior hyperparameters across different model sizes. This feature of REPS aligns with the first two reasons it is well-suited for our analysis: its ability to assess posterior sensitivity to prior assumptions and its feasibility across model sizes.

The REPS method solves the following problem:

$$\tilde{\lambda} = \min_{M(\lambda):M>0,\mathbb{E}_\pi=1} \mathbb{E}_p \left[\frac{M(\lambda) \cdot \lambda}{\mathbb{E}_p(M)} \right] \quad \text{s.t. } \mathbb{E}_\pi \left[M(\lambda) \log M(\lambda) \right] \leq \mathcal{R}.$$

Here, λ denotes the Minnesota tightness hyperparameter, $\pi(\lambda)$ is the original prior distribution, and $p(\lambda)$ denotes the corresponding posterior distribution. The alternative prior is defined as $\tilde{\pi}(\lambda) = M(\lambda)\pi(\lambda)$, where $M(\lambda)$ is a distortion function. We impose a constraint that ensures that the distance between the original prior and the alternative prior measured in relative entropy remains below a prespecified value $\mathcal{R}$, for which Ho (2023) provides a data-driven rule of thumb for choosing $\mathcal{R}$. Thus, the goal of REPS is to find the distortion function $M(\lambda)$ that minimizes the distorted posterior mean of λ, $\tilde{\lambda}$, subject to this entropy bound.

Ho (2023) shows that the solution to this problem, $M(\lambda)$, takes the simple exponential form:

$$M(\lambda) \propto \exp\left(\eta \cdot L(X|\lambda) \cdot (\lambda - \tilde{\lambda})\right)$$

where η is the Lagrange multiplier, $L(X|\lambda)$ is the likelihood of the data given λ, and $\tilde{\lambda}$ is the value of the objective function. Intuitively, the distortion function depends on the likelihood because greater distortion from using an alternative prior arises in regions where the likelihood is high.

By substituting the solution back into the objective function, it becomes:

$$\tilde{\lambda} = \mathbb{E}_p \left[\frac{\exp\left(\eta \cdot L(X|\lambda) \cdot (\lambda - \tilde{\lambda})\right)}{\mathbb{E}_p \left[\exp\left(\eta \cdot L(X|\lambda) \cdot (\lambda - \tilde{\lambda})\right) \right]} \lambda \right]$$

Given posterior draws of λ, denoted λ_i, this simplifies to:

$$\tilde{\lambda} \propto \sum_i \exp\left(L(X|\lambda_i) \cdot (\lambda_i - \tilde{\lambda})\right) \lambda_i$$

This formula shows that $\tilde{\lambda}$ is a weighted average of the posterior draws of λ, where the weights depend on the marginal likelihood. Note that $L(X|\lambda_i)$ corresponds to the marginal likelihood used for hyperparameter estimation in GLP, which is central to our analysis. By examining how the likelihood interacts with hyperparameter distortion, REPS sheds light on a key limitation of GLP: its overemphasis on in-sample fit through the likelihood. This feature of REPS corresponds to the third reason it is well-suited for our analysis—its ability to analyze how the likelihood shapes hyperparameter estimation.

To implement REPS, I first estimate the hyperparameter λ following GLP and obtain posterior draws, along with the corresponding marginal likelihood and prior density for each draw. Second, I solve the REPS minimization problem using these outputs to obtain the **distorted hyperparameter**, also referred to as the **REPS-based hyperparameter**. Third, I estimate the VAR model and generate point and density forecasts using both the posterior draws of λ from GLP and the REPS-based value of λ. Finally, I compare the resulting forecasts using the forecast metrics introduced earlier. This comparison allows me to assess whether the REPS-based hyperparameter improves forecast accuracy. Specifically, I explore improvements across different model sizes, forecast horizons, and time periods.

Algorithm:

1. Estimate λ following Giannone et al. (2015) and store the posterior draws of the hyperparameters, along with the corresponding marginal likelihood and the prior density for each draw.
2. Using the outputs from Step 1, apply REPS to obtain the distorted values of λ.
3. Estimate the VAR model and generate forecasts using the posterior draws of λ from Giannone et al. (2015) and REPS-based λ, respectively.
4. Compare the resulting point and density forecasts using forecast metrics.

In addition to the baseline comparison, I conduct two complementary robustness exercises in the Appendix. First, I replace the GLP posterior draws of λ with their posterior mean and compare the resulting GLP plug-in forecasts with the REPS plug-in forecasts. This exercise removes hyperparameter uncertainty from the GLP benchmark and isolates the role of the selected value of λ. Second, I implement REPS as a posterior reweighting scheme. In this exercise, I retain the GLP posterior draws of λ and the associated predictive distributions, but replace the equal mixture weights used by GLP with weights implied by the REPS distortion function. This preserves the GLP posterior-mixture structure and examines whether the REPS distortion has predictive content beyond the use of a single plug-in hyperparameter.

5 Data

This paper uses a macroeconomic dataset for the US economy, FRED-QD, constructed by McCracken and Ng (2016) and maintained by the Federal Reserve Bank of St.Louis. The dataset covers the period from 1959Q1 to 2023Q4 and consists of 248 quarterly series. Since the variable Capacity utilization is only available starting from 1967Q1, my sample begins from that period. I take the log difference and multiply by 400 for all variables except the unemployment rate and the federal funds rate, to make the variables approximately stationary and easier to interpret.

Table 2. Variables and Transformation by Model Size

Variables	**FRED Mnemonic**	**Transformation**	**VAR**	
			Small	**Medium**
Real GDP	GDPC1	400×log-diff	x	x
GDP deflator	GDPCTPI	400×log-diff	x	x
Unemployment rate	UNRATE	level	x	x
Federal funds rate	FEDFUNDS	level	x	x
Real consumption	PCECC96	400×log-diff		x
Real investment	GPDIC1	400×log-diff		x
Hours worked: Business sector	HOANBS	400×log-diff		x
Real compensation per hours	CPMPRNFB	400×log-diff		x
Consumer price index	CPIAUCSL	400×log-diff		
Commodity price	CUSR0000SAC	400×log-diff		
Industrial production	INDPRO	400×log-diff		
Employment	CE160V	400×log-diff		
Employment in the services sector	SRVPRD	400×log-diff		
Real residential investment	PRFIx	400×log-diff		
Nonresidential investment	PNFIx	400×log-diff		
Personal consumption expenditures, price index	PCECTPI	400×log-diff		
Gross private domestic investment, price index	GPDICTPI	400×log-diff		
Capacity utilization	TCU	400×log-diff		
Consumer expectations	UMCSENTx	400×log-diff		
One-year bond rate	GS1	400×log-diff		
Five-years bond rate	GS5	400×log-diff		
S&P500	S&P 500	400×log-diff		
Trade weighted US dollar index	TWEXAFEGSMTHx	400×log-diff		
Real M2	M2SL	400×log-diff		

6 Results

6.1 Point Forecast Accuracy

Figure 1 plots the percentage change in RMSFE from using the REPS-based λ relative to the posterior draws of λ from GLP. Improvements in point forecast accuracy from using REPS in small- and medium-sized VARs, especially as the forecast horizon increases in all periods except during COVID. In large VAR, the gains are concentrated in persistent variables, particularly the unemployment rate and the federal funds rate. During COVID, REPS outperforms GLP primarily in small-sized VARs.

To summarize, improvements in point forecast accuracy from using REPS are noticeable for:

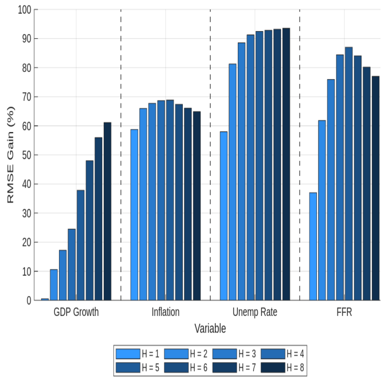
1. **Model size:** Across all model sizes
2. **Horizon:** Longer forecast horizons

3. **Period:** Small-sized VARs during the COVID period

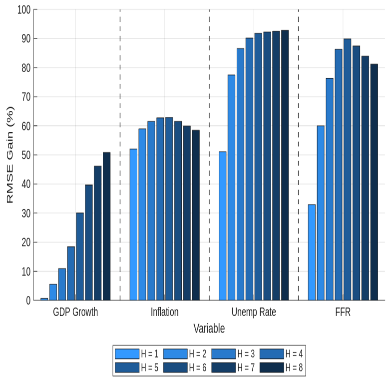
Figure 1: Percentage Change in RMSFE

1. Sample 1

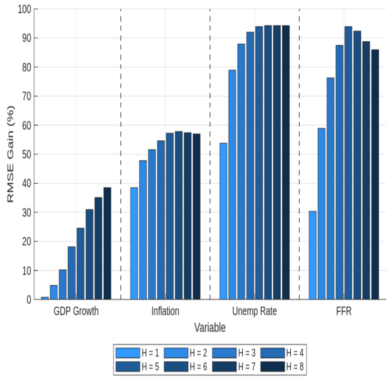
Small VAR



Medium VAR

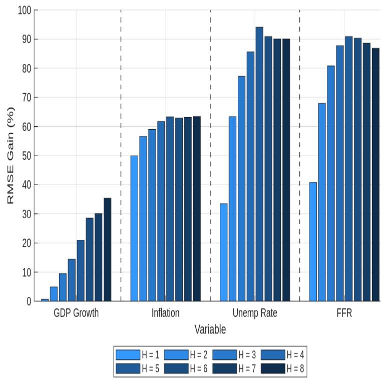


Large VAR

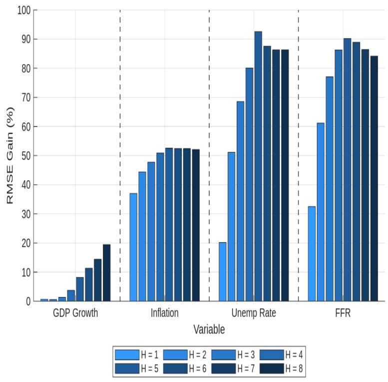


2. Sample 2

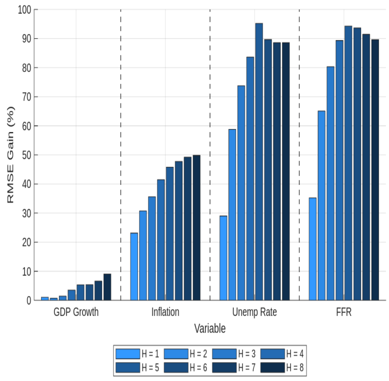
Small VAR



Medium VAR

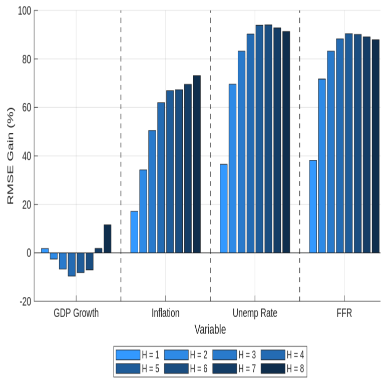


Large VAR

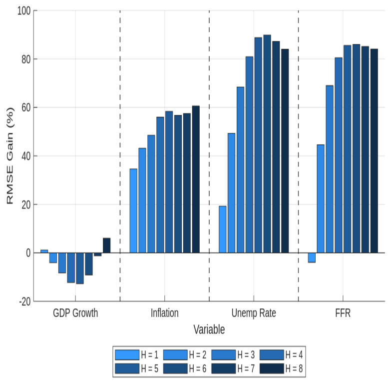


3. Financial Crisis

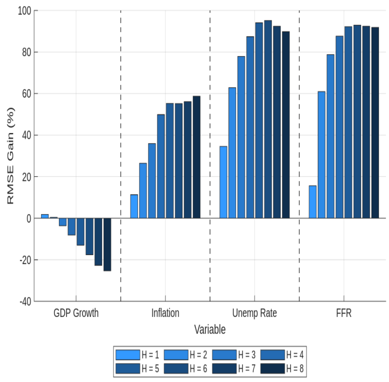
Small VAR



Medium VAR

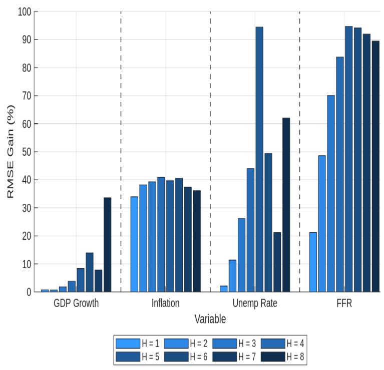


Large VAR

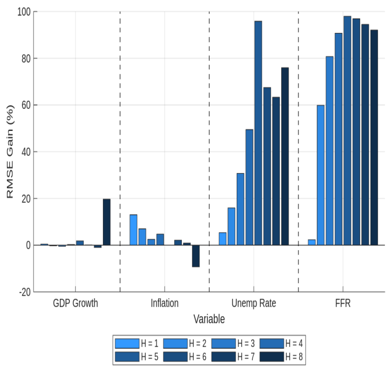


4. COVID

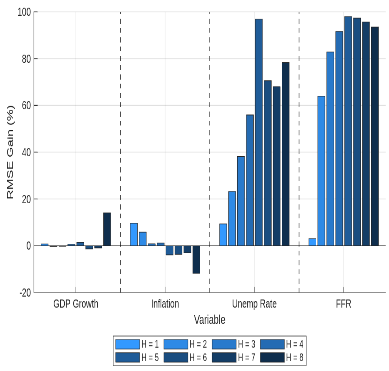
Small VAR



Medium VAR



Large VAR



Tests of equal forecast accuracy using the Diebold and Mariano (1995) and Clark West (2007) tests show that different hyperparameter choices lead to statistically significant differences in forecast performance in most variable-hofizon combinations across all three model sizes. Therefore, hyperparameters obtained through REPS not only improve point forecasts but also produce forecasts that are statistically distinct from those generated using the approach of GLP.

Table 3. Sample 1: p-values from DM test and CW test

Horizon	1	2	3	4	5	6	7	8
Small								
DM test								
- GDP growth rate	0.1763	0.0180	0.0002	0.0000	0.0000	0.0000	0.0000	0.0000
- Inflation	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.6215	0.4693
- Unemployment rate	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000
- Federal funds rate	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000
CW test								
- GDP growth rate	0.1202	0.0257	0.0261	0.0159	0.0044	0.0008	0.0002	0.0000
- Inflation	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000
- Unemployment rate	0.0001	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000
- Federal funds rate	0.0008	0.0015	0.0028	0.0026	0.0015	0.0008	0.0004	0.0002
Medium								
DM test								
- GDP growth rate	0.0630	0.0280	0.0068	0.0013	0.0000	0.0000	0.0000	0.0000
- Inflation	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000
- Unemployment rate	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000
- Federal funds rate	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000
CW test								
- GDP growth rate	0.0747	0.0694	0.0099	0.0001	0.0000	0.0000	0.0000	0.0000
- Inflation	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000
- Unemployment rate	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000
- Federal funds rate	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000
Large								
DM test								
- GDP growth rate	0.0280	0.0425	0.0630	0.0013	0.0002	0.0000	0.0000	0.0000
- Inflation	0.0007	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000
- Unemployment rate	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000
- Federal funds rate	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000
CW test								
- GDP growth rate	0.0084	0.0294	0.0096	0.0002	0.0001	0.0004	0.0006	0.0001
- Inflation	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000
- Unemployment rate	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000
- Federal funds rate	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000

These gains raise the following questions: Why does forecast accuracy improve when using the REPS-based λ instead of the posterior estimates obtained from GLP? Why do the improvements vary across different periods and model sizes? To address these questions, we analyze how REPS reshapes the trade-off between in-sample fit and forecast accuracy embedded in the likelihood.

6.2 Trade-off Between In-Sample Fit and Forecast Accuracy: REPS vs GLP

Recall that GLP use the following marginal likelihood, expressed as a function of the hyperparameter, as the likelihood function for hyperparameter estimation:

$$p(y|\lambda) \propto \underbrace{\left|\left(V_{\varepsilon}^{\text{posterior}}\right)^{-1} V_{\varepsilon}^{\text{prior}}\right|^{\frac{T-p+d}{2}}}_{\text{In-sample fit}} \cdot \underbrace{\prod_{t=p+1}^{T} \left|V_{t|t-1}\right|^{-\frac{1}{2}}}_{\text{Forecast accuracy}}$$

Here, λ denotes the Minnesota prior tightness hyperparameter, and $V_{t|t-1} = \mathbb{E}_{\Sigma}[\text{Var}(y_t|y^{t-1}, \Sigma)]$ is the variance of the one-step-ahead forecast of y_t (conditional on Σ), averaged over all prior realizations of Σ.

They claim this method balances in-sample fit and forecast accuracy. However, it places excessive weight on in-sample fit, therefore may be ill-suited for forecasting purposes. As a result, the estimated hyperparameter may overfit the estimation sample, leading to poor performance in out-of-sample forecasts. Instead, an alternative hyperparameter choice can improve out-of-sample forecasts, though at the expense of in-sample fit. This is precisely what REPS-based λ achieves.

To better understand how REPS reshapes the balance between in-sample fit and forecast accuracy compared to GLP, I compute each component of the marginal likelihood and evaluate how the REPS-based λ affects these components relative to the λ chosen by GLP. After taking logarithms, the marginal likelihood can be rewritten as:

$$\log p(y|\gamma) \propto \underbrace{\frac{T-p+d}{2} \cdot \log\left|\left(V_{\varepsilon}^{\text{posterior}}\right)^{-1} V_{\varepsilon}^{\text{prior}}\right|}_{\textbf{In-sample-fit} \downarrow} - \frac{1}{2} \underbrace{\sum_{t=p+1}^{T} \log\left|V_{t|t-1}\right|}_{\textbf{Forecast Accuracy} \uparrow}$$

where T is the window size, p is the lag order, and $d = N + 2$ is the degrees of freedom in the inverse-Wishart prior for Σ. I compute each component separately under both methods and calculate the percentage change to assess how REPS shifts the balance between in-sample fit and forecast accuracy.

Table 4. Change In-sample Fit(%) $\left(\frac{\textbf{REPS}}{\textbf{GLP}}\right)$

Periods	Small	Medium	Large
Sample 1	-0.6202	-0.4138	-0.2165
Sample 2	-0.6746	-0.4359	-0.2244
Financial Crisis	-0.7051	-0.4906	-0.2447
COVID	-0.8519	-0.2831	-0.1704

Table 5. Change in Forecast Accuracy(%) $\left(\frac{\textbf{REPS}}{\textbf{GLP}}\right)$

Periods	Small	Medium	Large
Sample 1	0.6930	0.4734	0.2839
Sample 2	0.7919	0.5163	0.2836
Financial Crisis	0.9054	0.5841	0.2841
COVID	0.8986	0.3747	0.2844

Tables 4 and 5 show that REPS consistently yields substantial improvements in forecast accuracy, at the cost of a moderate reduction in in-sample fit, relative to GLP. These changes in in-sample fit and forecast accuracy are more pronounced in small- and medium-sized VARs in Sample 1, Sample 2, and during the financial crisis, and in small-sized VARs during the COVID period. These patterns are consistent with the results in Figure 1, which indicate that gains in forecast performance are most pronounced for small- and medium-sized VARs. In large VARs, consistent with limited likelihood-side adjustment, forecast gains are less broad-based and are concentrated in persistent variables. This suggests that REPS enhances forecast accuracy by shifting the balance between in-sample fit and forecast accuracy embedded in the likelihood.

Two features of REPS help address a key limitation of Giannone et al's approach: its tendency to overfit the estimation sample. First, REPS mitigates this issue by adjusting λ more aggressively when the marginal likelihood is high, as discussed in Section 4. In other words, for values of λ that yield better in-sample fit, REPS introduces greater distortion, making the selected hyperparameter less constrained by in-sample fit. Second, because REPS solves a minimization problem[1] over prior distortions, it typically selects a value of λ that is lower than the posterior mean estimated under GLP, thereby imposing stronger prior shrinkage. This makes the resulting model less reliant on the estimation sample and better suited for out-of-sample forecasting.

Still, one question remains. Why do forecast accuracy gains vary across different model sizes and periods?

6.3 REPS and the Marginal Likelihood

The extent to which REPS improves forecast accuracy depends on how the distorted hyperparameter interacts with the marginal likelihood.

First, the marginal likelihood governs how much REPS adjusts the hyperparameter. Table 6 shows the percentage difference between the REPS-based λ and the posterior mean estimated under GLP. The REPS-based λ deviates substantially from the posterior mean in small and medium VARs and during COVID. In all sample periods except during COVID, greater distortion is associated with larger improvements in forecast accuracy. However, COVID is an exception: despite substantial distortions across all model sizes, the forecast accuracy improvements in medium and large VARs remain limited. This directly leads to the second point.

Table 6. Percentage Change in λ

$$\frac{\lambda_{\text{REPS}} - \lambda_{\text{mean, GLP}}}{\lambda_{\text{mean, GLP}}} \times 100$$

	Small	Medium	Large
Sample 1	-18.01	-14.29	-8.51
Sample 2	-19.48	-16.08	-10.50
Financial Crisis	-19.01	-15.93	-35.03
COVID	-34.80	-30.15	-27.15

Second, once λ is distorted, the marginal likelihood also determines the trade-off—how much in-sample fit decreases, and how much forecast accuracy improves. Specifically, in all sample periods except during COVID, large distortions in λ lead to substantial changes in both in-sample fit and forecast accuracy due to the dispersed shape of the likelihood. In contrast, during COVID, these changes are limited because the likelihood is elevated and relatively flat.

To examine these two mechanisms more closely, we now return to the REPS problem. Recall that given the posterior draws of λ, REPS-based λ can be expressed as follows:

$$\tilde{\lambda} \propto \sum_i \underbrace{\exp\left[L(X|\lambda_i) \cdot (\lambda_i - \tilde{\lambda})\right]}_{\text{Weight}} \cdot \lambda_i$$

The distortion in λ is amplified in two cases: when many λ draws fall below $\tilde{\lambda}$ with nonzero likelihood, and when the marginal likelihood is high in that regions. In these cases, the weighting puts more weight on smaller λ values, pulling λ downward.

[1]Alternatively, one can solve a maximization problem in REPS, which is likely to yield a value of λ higher than the posterior mean estimated under GLP, imposing less shrinkage and letting the data speak more—thereby increasing in-sample fit but potentially reducing forecast accuracy.

$$\tilde{\lambda} \propto \sum_i \exp\left[\underbrace{L(X|\lambda_i)}_{(1)\text{ many }(2)\text{ high}} \cdot \underbrace{(\lambda_i - \tilde{\lambda})}_{\text{negative}}\right] \cdot \lambda_i$$

1. Many $\lambda_i < \tilde{\lambda}$ with nonzero $L(X|\lambda_i)$.
2. $L(X|\lambda_i)$ is high when $\lambda_i < \tilde{\lambda}$.

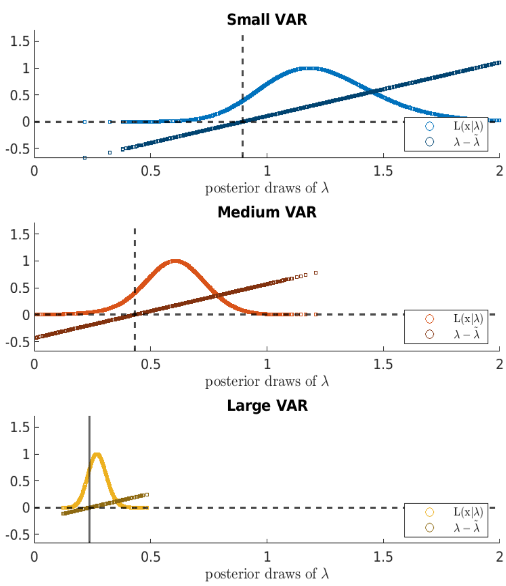
Each condition corresponds to a distinct feature of the marginal likelihood: the first reflects its dispersion while the second captures its magnitude. Therefore, to understand the extent of hyperparameter distortion in REPS, it is essential to examine the shape of the marginal likelihood. The degree of distortion depends on:

1. The **dispersion** of the marginal likelihood function: If many posterior draws of λ fall below $\tilde{\lambda}$, the distortion increases.
2. The **magnitude** of the marginal likelihood function: If the marginal likelihood value itself is particularly high at specific λ values, the distortion becomes more pronounced.

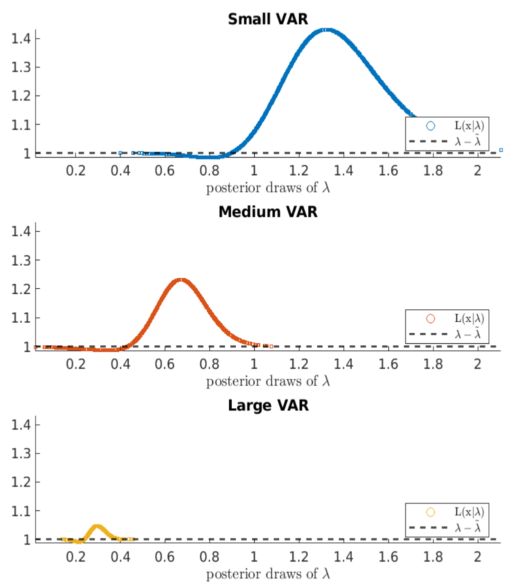
What does the shape of the marginal likelihood reveal? The marginal likelihood essentially measures how well different values of hyperparameters explain the data, and its shape has the following implications. A dispersed likelihood means different parameter values explain the data similarly well. A highly peaked likelihood indicates that a certain set of parameter values have significantly more explanatory power than others. (**Dispersion**) A high likelihood value at particular λ values means those values explain the data significantly better than others. (**Magnitude**)

To examine how the shape of the marginal likelihood interacts with REPS, Figure 2 plots the marginal likelihood, $L(X|\lambda)$, the deviation term, $(\lambda - \tilde{\lambda})$, and the corresponding weighting function , $\exp\left(L(X|\lambda_i)(\lambda_i - \tilde{\lambda})\right)$, across model sizes.

Figure 2: Drivers of Hyperparameter Distortion in REPS by Model Size



(a) Distribution of $L(X|\lambda)$ and $(\lambda - \tilde{\lambda})$



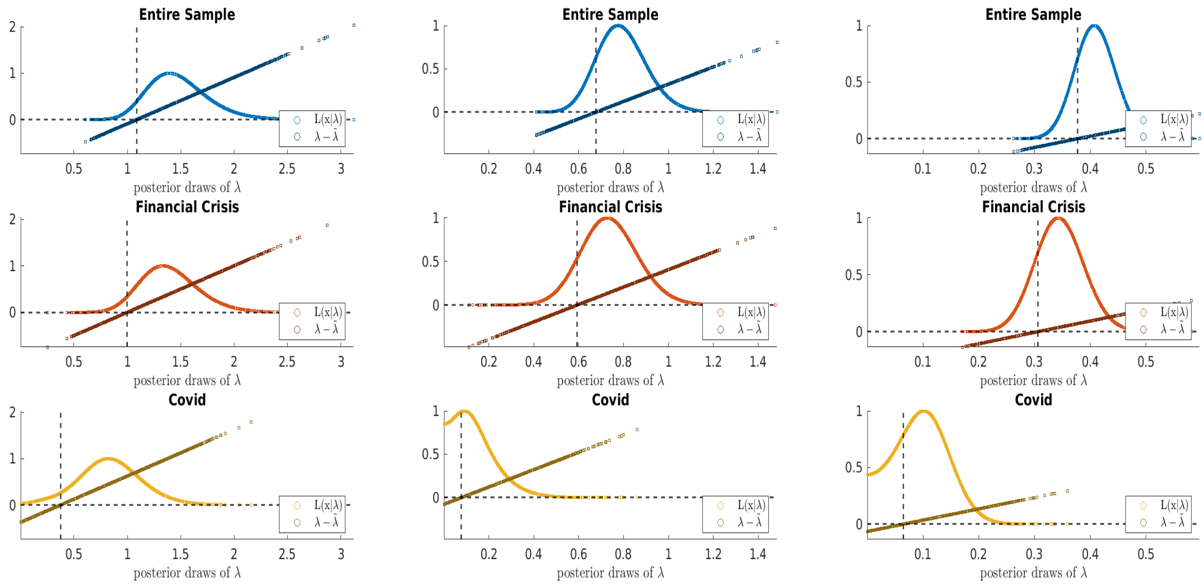
(b) Distribution of $\exp\left(L(X|\lambda) \cdot (\lambda - \tilde{\lambda})\right)$

Note: Horizontal line $y = 0$, vertical line $\lambda = \tilde{\lambda}$. The figure is drawn for the 70th rolling window in Sample 1.

Dispersion of the Likelihood and Model Size: In small VAR models, the marginal likelihood is more dispersed, meaning a wider range of λ values explain the data similarly well (Panel (a)). This is because small models have fewer coefficients and therefore require less aggressive shrinkage. Consequently, many λ draws lie below $\tilde{\lambda}$, and the likelihood surface flattens out. Accordingly, the weight term often falls below one (Panel (b)), leading to greater distortion under REPS. As REPS induces substantial shifts in the hyperparameter, the resulting change in the marginal likelihood is large–yielding significant improvements in out-of-sample forecast accuracy at the cost of a moderate reduction in in-sample fit. This mechanism explains the significant forecast improvements observed in small-sized VARs.

In contrast, large VARs demand stronger shrinkage, resulting in sharply peaked likelihoods centered around low λ values (Panel (a)). As a result, REPS induces only minor adjustments in the hyperparameter. This leads to small changes in the marginal likelihood, and thus the changes in in-sample fit and forecast accuracy remain limited. This mechanism explains the small forecast improvements observed in large-sized VARs.

Figure 3: Drivers of Hyperparameter Distortion in REPS by Period



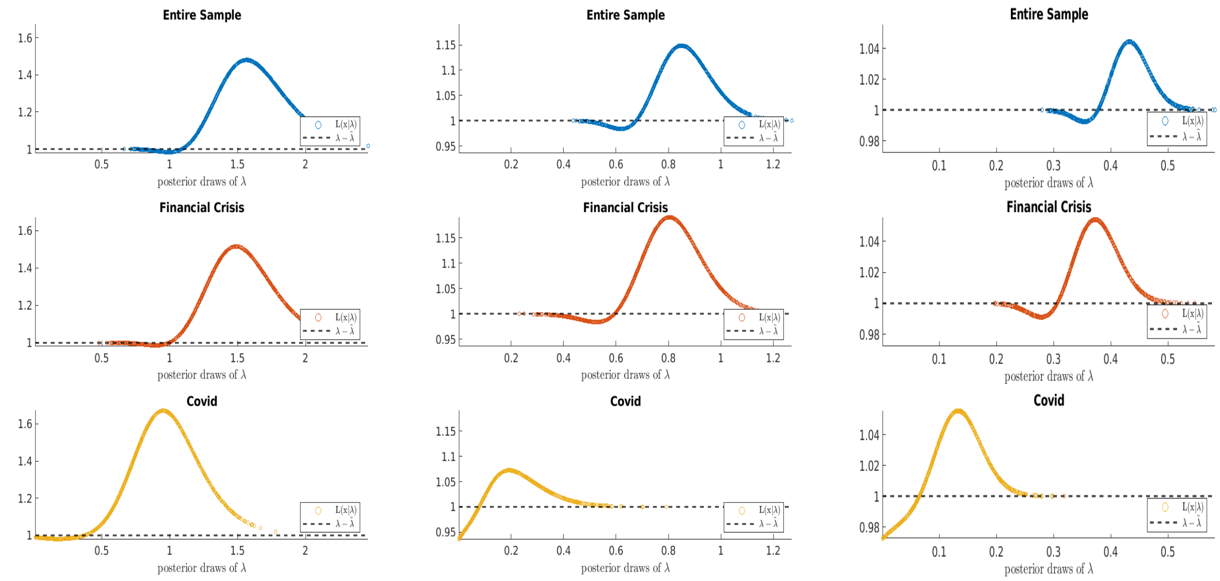
(a) Small VAR (b) Medium VAR (c) Large VAR

Note: Horizontal line $y = 0$, vertical line $\lambda = \tilde{\lambda}$. The figure is drawn for the 36, 85, and 134th rolling window in Sample 1, the financial crisis, and COVID respectively.

Magnitude of the Likelihood and Time Periods: To analyze how the interaction between the marginal likelihood and REPS varies across time periods, Figure 3 shows the marginal likelihood, $(L(X|\lambda))$, and the deviation term, $(\lambda - \tilde{\lambda})$, across different time periods. Figure 4 presents the corresponding weighting function, $\exp\Big(L(X|\lambda_i)(\lambda_i - \tilde{\lambda})\Big)$.

Notably, during COVID, the likelihood is particularly high over a broad range of small λ draws in medium and large VARs (Figure 3). This occurs because the model relies more heavily on prior information rather than capturing irregularities in the data. As a result, the weights on many λ draws, $\exp\Big(L(x|\lambda_i)(\lambda_i - \tilde{\lambda})\Big)$, fall well below one (Figure 4). As a result, REPS induces a large distortion in the hyperparameter. However, because the likelihood surface is relatively flat in that region, the resulting change in likelihood and the corresponding changes in in-sample fit and forecast accuracy remain limited. This mechanism explains limited forecast accuracy improvements observed in medium and large VARs during COVID.

Figure 4: Drivers of Hyperparameter Distortion in REPS by Period – Weight



(a) Small VAR (b) Medium VAR (c) Large VAR

Note: The figure is drawn for the 36, 85, and 134th rolling window in Sample 1, the financial crisis, and COVID respectively.

Multi-step ahead forecasting: Lastly, compared to the hyperparameter estimates from GLP, the REPS-based λ consistently yields greater improvements at longer forecast horizons. This is because the marginal likelihood places excessive weight on in-sample fit, which can hinder out-of-sample performance. By relaxing this constraint, REPS enables more flexible and effective long-term forecasting.

Conjecture 1 (Size): The REPS-based hyperparameter improves forecast accuracy in small- and medium-sized VARs, where the marginal likelihood is more dispersed. However, the gains diminish in large VARs, where the marginal likelihood is sharply peaked, limiting changes in both in-sample fit and forecast accuracy.

Conjecture 2 (Time Periods): The REPS-based hyperparameter improves forecast accuracy in most periods, except during COVID, where the exceptionally elevated and flat likelihood limits gains despite substantial shifts in the hyperparameter.

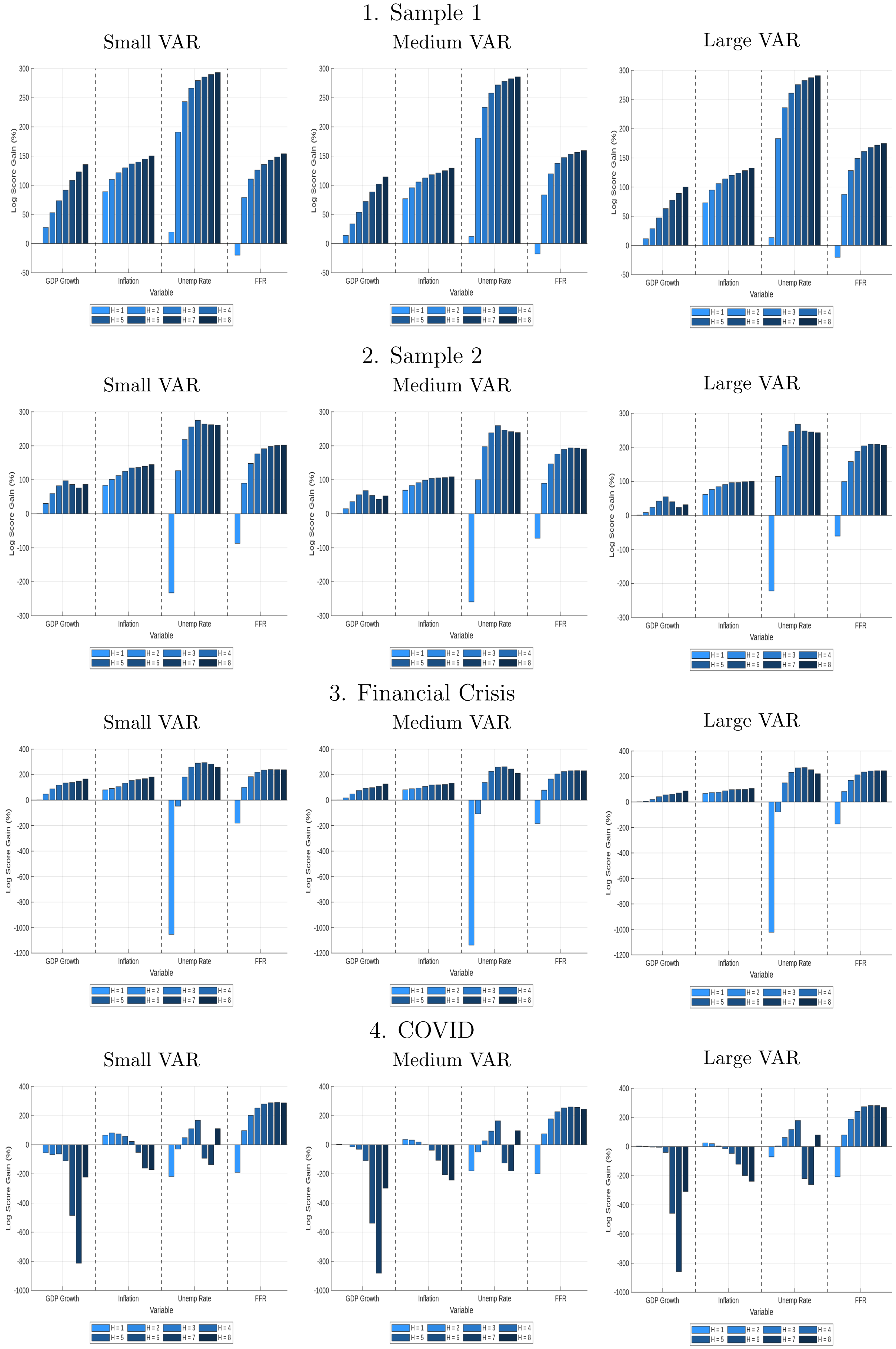
Conjecture 3 (Horizons): By relaxing the in-sample fit constraint embedded in the marginal likelihood, the REPS-based hyperparameter enhances forecasting accuracy over longer horizons.

6.4 Density Forecast Accuracy

Since the REPS-based λ is typically smaller than the posterior mean estimated under GLP, the distorted λ reduces the prior variance on the VAR coefficients and imposes stronger shrinkage. Consequently, the posterior distribution of the coefficients becomes more concentrated, leading to more sharply peaked density forecasts under REPS.

When forecasts are accurate—as in the case of large VAR models—this increased concentration enhances the log score. However, when forecasts are less accurate, particularly during COVID, overly peaked density forecasts can lead to a substantial reduction in the log score due to overconfidence in incorrect predictions. In Figure 5, there is a substantial loss in log score during the COVID period.

Figure 5: Percentage Change in Log Score



To illustrate this more clearly, note that the log predictive score, $s_t(y^i_{t+h})$, is computed based on the quadratic approximation for Gaussian predictive distribution,

$$s_t(y^i_{t+h}) = -0.5\left[\ln(2\pi) + \ln(V^i_{t+h|t}) + \frac{(y^i_{t+h} - \overline{y}^i_{t+h|t})^2}{V^i_{t+h|t}}\right]$$

where, given F_{draw} from the forecast distribution of y^i_{t+h}, $\left\{y^i_{t+h,f}\right\}_{f=1}^{F_{draw}}$, $\overline{y}^i_{t+h|t}$is the posterior mean of the simulated forecast distributions, $V^i_{t+h|t}$ is the posterior variance of the simulated forecast distributions, and y^i_{t+h} is the actual value. Under REPS, $V^i_{t+h|t}$ tends to be much smaller than under the method of Giannone et al. As a result, when the squared forecast error $(y^i_{t+h} - \overline{y}^i_{t+h|t})^2$ is positive—that forecasts are imprecise—the negative impact on the log score is more severe under Ho's method due to its tigher predictive distribution.

Tests of equal forecast accuracy using the Amisano and Giacomini (2007) test show that different hyperparameter choices lead to statistically significant differences in forecast performance (Table 7). Therefore, hyperparameters obtained through REPS produce density forecasts that are statistically distinct from those generated using the approach of GLP.

Table 7. Sample 1: p-values from AG test

Horizon	1	2	3	4	5	6	7	8
Small								
- GDP growth rate	0.4438	0.0000	0.0000	0.0000	0.0000	0.0102	0.0981	0.0639
- Inflation	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000
- Unemployment rate	0.0160	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000
- Federal funds rate	0.0737	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000
Medium								
- GDP growth rate	0.2166	0.0000	0.0000	0.0000	0.0000	0.0974	0.3479	0.2597
- Inflation	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000
- Unemployment rate	0.0131	0.0005	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000
- Federal funds rate	0.0760	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000
Large								
- GDP growth rate	0.0047	0.0000	0.0000	0.0000	0.0000	0.1652	0.5919	0.4851
- Inflation	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000
- Unemployment rate	0.0186	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000
- Federal funds rate	0.1231	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000

7 Conclusion

Selecting appropriate shrinkage priors is crucial for forecasting with parameter-rich VAR models. However, the method proposed by Giannone et al. (2015) faces two key limitations: (i) sensitivity to the choice of hyperprior, and (ii) excessive weight on in-sample fit in the likelihood. To address these issues, this paper employs Relative Entropy Prior Sensitivity (REPS), which is particularly well-suited to our analysis due to (i) its applicability to models of any size, and (ii) its ability to assess the role of the likelihood in hyperparameter estimation. I find that REPS-based hyperparameters alleviate the limitations of Giannone et al. (2015) and improve point forecast accuracy particularly: (i) in small- and medium-sized VARs; (ii) at longer forecast horizons; (iii) in small-sized VARs during COVID. These improvements are statistically significant. I also find that these patterns are attributed to the shape of the likelihood. The findings underscore the importance of prior selection in VAR forecasting across different model sizes and periods. Moreover, the paper provides a systematic framework for enhancing prior selection in Bayesian VAR forecasts.

There are several avenues for future research. First, one could incorporate alternative priors and examine how REPS-based hyperparameters perform across different prior structures. A natural extension would be to include a cross-variable shrinkage hyperparameter in addition to the overall tightness parameter λ. This would allow REPS to account for both overall and cross-variable shrinkage in the Minnesota prior. Furthermore, while this paper uses the hyperparameter itself in REPS's objective function, the objective could be tailored more directly to forecasting performance. For example, one could use the RMSFE as the criterion in the REPS procedure, thereby selecting hyperparameters that directly minimize forecast errors. This approach may offer more targeted gains in predictive accuracy, though it would likely introduce additional computational complexity. Lastly, given the relatively modest improvements in VAR forecast performance during the COVID period, future research could focus on developing hyperparameter selection methods that enhance accuracy even in periods of extreme volatility.

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Appendices

A Alternative Forecast Specifications

On the main analysis, I evaluate whether using a single REPS-based value of the Minnesota tightness parameter, $\tilde{\lambda}$, improves forecast accuracy relative to the standard GLP approach, which generates forecasts by integrating over posterior draws of λ. This comparison evaluates the practical REPS plug-in forecasting rule proposed in this paper. However, it is not a pure comparison of hyperparameter values, because the two approaches differ in their treatment of hyperparameter uncertainty: GLP averages over posterior draws of λ, whereas the REPS plug-in forecast conditions on a single distorted value.

To assess whether the baseline results reflect the selected value of λ or the different treatment of λ-uncertainty, I conduct two complementary robustness exercises. First, I replace the GLP posterior draws of λ with their posterior mean and compare this GLP plug-in forecast with the REPS plug-in forecast (Appendix A.1.). This comparison places both methods on equal footing as plug-in forecasts and isolates the role of the selected hyperparameter value. Second, I implement REPS as a posterior reweighting scheme (Appendix A.2.) In this exercise, I retain the GLP posterior draws of λ, but replace the equal mixture weights used by GLP with REPS distortion weights. This preserves the posterior-mixture structure of GLP and examines whether the REPS distortion has predictive content beyond the use of a single plug-in hyperparameter.

A.1 Forecast Accuracy Improvements Using Posterior Mean Instead of Posterior Draws of the Hyperparameter

First, I use the posterior mean of λ. In this case, the comparison between GLP and REPS reflects only the uncertainty arising from the VAR coefficients, whereas using posterior draws captures uncertainty from both the VAR coefficients and the hyperparameter. Consequently, forecast accuracy improvement from REPS is expected to decrease, as the GLP forecasts no longer reflect hyperparameter variation.

This is reflected in Figure 7: point forecast accuracy improvements from REPS relative to GLP decrease significantly across all model sizes and periods. Notably, the unemployment rate forecasts consistently worsen. Still, in Sample 1 and 2, forecasts improve with the use of the REPS-based λ for all variables except the unemployment rate in small and medium VARs, and these improvements are statistically significant (Table 8).

Figure 7: Percentage Change in RMSFE

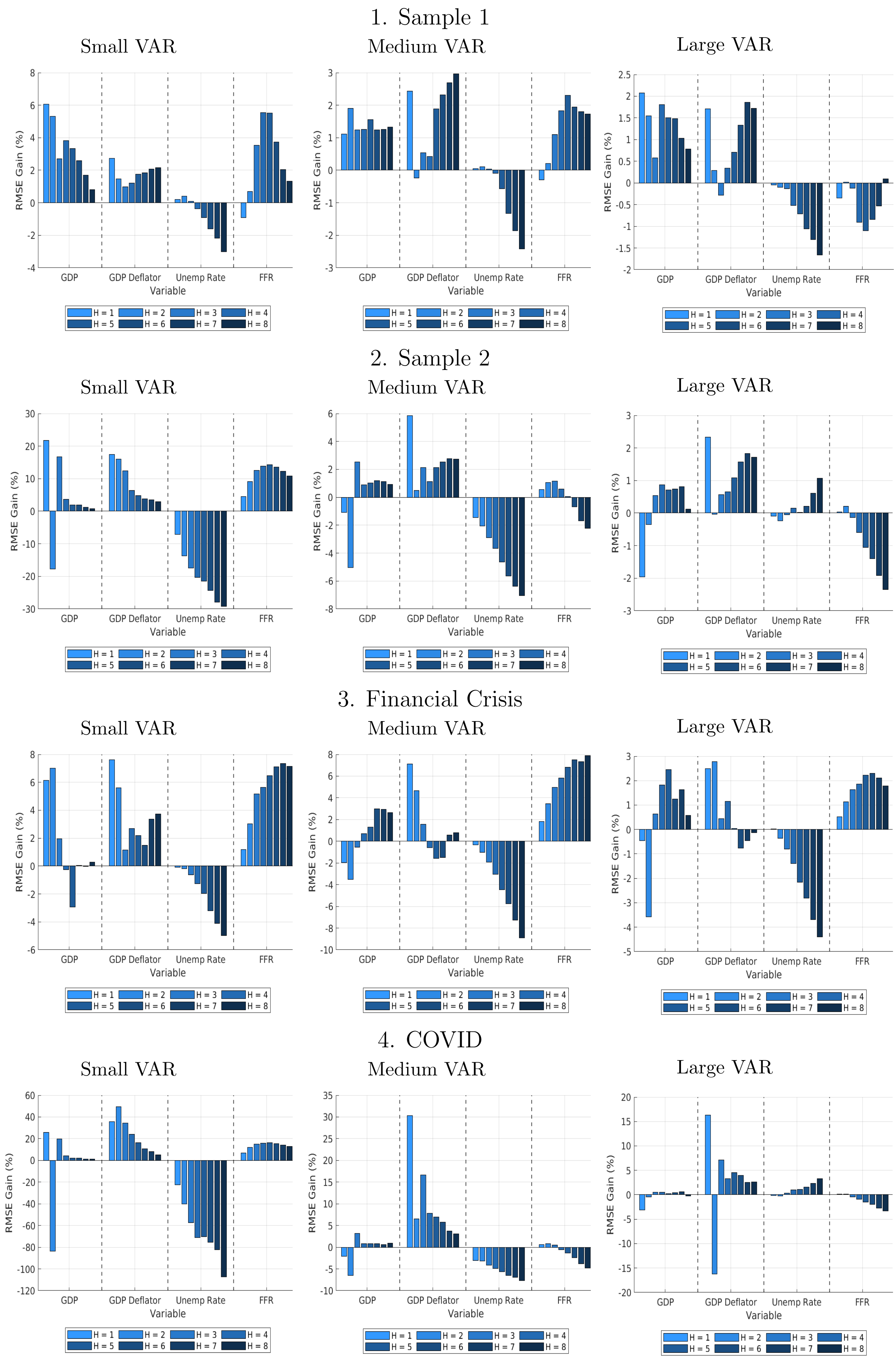


Table 8. Sample 1: p-values from DM test and CW test

Horizon	1	2	3	4	5	6	7	8
Small								
DM test								
- GDP growth rate	0.0027	0.0134	0.0202	0.0031	0.0002	0.0015	0.0244	0.2824
- Inflation	0.0137	0.2873	0.4977	0.2838	0.0816	0.0509	0.0131	0.0094
- Unemployment rate	0.2468	0.2020	0.8744	0.6413	0.3535	0.2184	0.1301	0.1120
- Federal funds rate	0.1174	0.4646	0.0085	0.0008	0.0021	0.0512	0.3783	0.5501
CW test								
- GDP growth rate	0.0003	0.0054	0.0039	0.0011	0.0001	0.0003	0.0068	0.0965
- Inflation	0.003	0.1017	0.1988	0.1117	0.032	0.0207	0.0053	0.0039
- Unemployment rate	0.1113	0.0871	0.4064	0.6495	0.7997	0.8748	0.9227	0.9333
- Federal funds rate	0.9045	0.1218	0.0008	0.0001	0.0002	0.0053	0.0865	0.1576
Medium								
DM test								
- GDP growth rate	0.5172	0.0654	0.1098	0.0651	0.0038	0.0455	0.0103	0.0178
- Inflation	0.0326	0.855	0.4998	0.5779	0.0024	0.0041	0.008	0.0115
- Unemployment rate	0.5318	0.3834	0.8500	0.7208	0.0179	0.0001	0.0000	0.0000
- Federal funds rate	0.5624	0.6610	0.0889	0.0159	0.0184	0.1187	0.1930	0.1604
CW test								
- GDP growth rate	0.0936	0.0249	0.0409	0.0253	0.0011	0.0171	0.0037	0.0069
- Inflation	0.0066	0.5023	0.2103	0.2515	0.001	0.0018	0.0036	0.0055
- Unemployment rate	0.2556	0.1831	0.4057	0.6176	0.9891	1.0000	1.0000	1.0000
- Federal funds rate	0.6751	0.2796	0.0311	0.0042	0.0042	0.0310	0.0562	0.0443
Large								
DM test								
- GDP growth rate	0.0466	0.0030	0.2279	0.0104	0.0067	0.0449	0.0814	0.1360
- Inflation	0.1104	0.6696	0.7138	0.5144	0.1030	0.0001	0.0001	0.0005
- Unemployment rate	0.5807	0.1895	0.2047	0.0016	0.0000	0.0003	0.0006	0.0002
- Federal funds rate	0.2540	0.9532	0.7603	0.3108	0.2652	0.4228	0.6494	0.9343
CW test								
- GDP growth rate	0.0045	0.0009	0.0842	0.0041	0.0025	0.0191	0.0333	0.0565
- Inflation	0.0412	0.2909	0.6072	0.2251	0.0411	0.0000	0.0001	0.0002
- Unemployment rate	0.7009	0.9004	0.8893	0.9991	1.0000	0.9998	0.9996	0.9999
- Federal funds rate	0.8455	0.4279	0.5515	0.7937	0.7881	0.6223	0.4223	0.2246

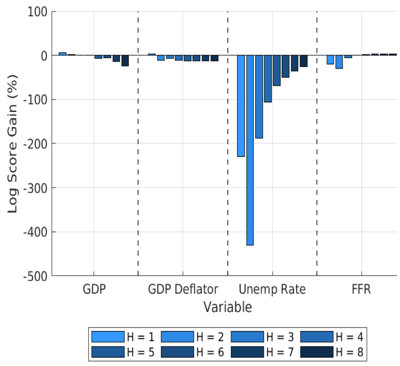
For density forecasts, the reduction in uncertainty leads to lower log score losses from using the REPS-based λ (Figure 8). Recall that REPS-based λ generates sharply peaked forecast densities, whereas posterior draws of λ yield more dispersed forecast distributions. This dispersion can lead to higher log scores for GLP-based λ when the point forecast is relatively inaccurate, as the broader distribution assigns non-negligible probability to a wider range of outcomes.

By instead using the posterior mean of λ, hyperparameter uncertainty is removed. As a result, GLP also produces a sharply peaked forecast distribution, reducing the relative disadvantage of REPS in terms of log score.

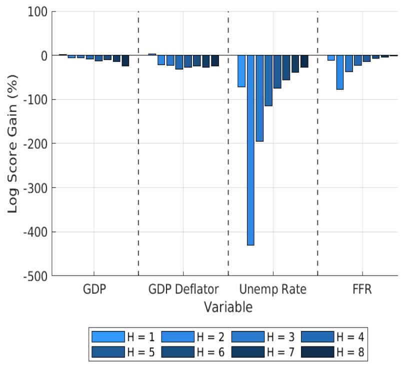
Figure 8: Percentage Chane in Log Score

1. Sample 1

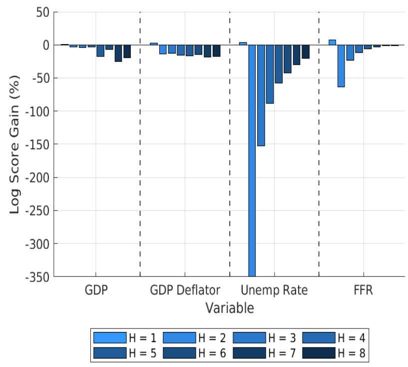
Small VAR



Medium VAR

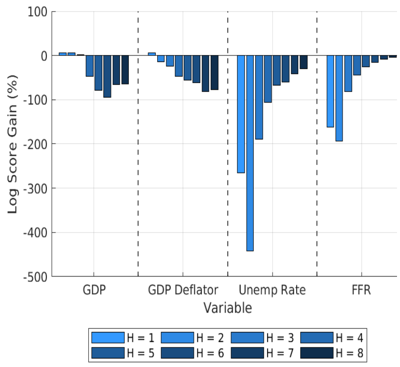


Large VAR

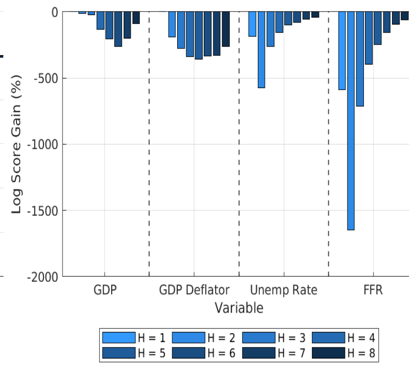


2. Sample 2

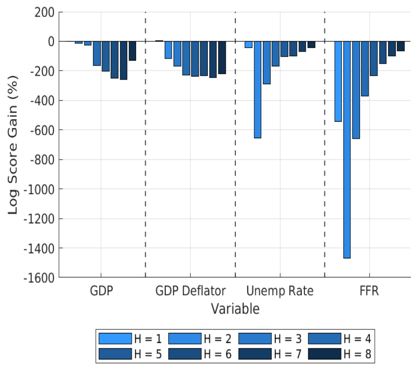
Small VAR



Medium VAR

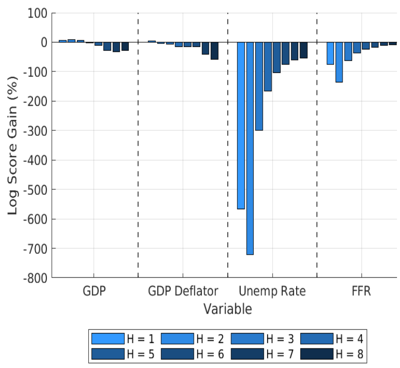


Large VAR

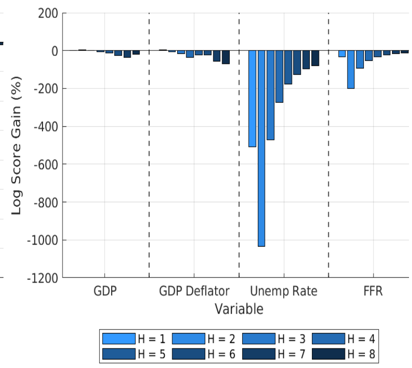


3. Financial Crisis

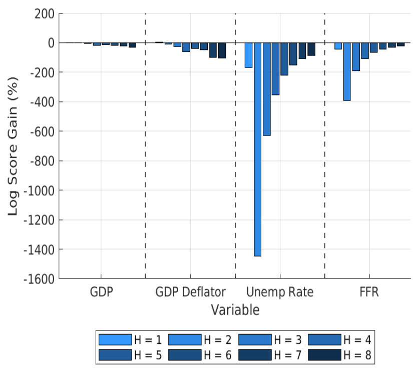
Small VAR



Medium VAR

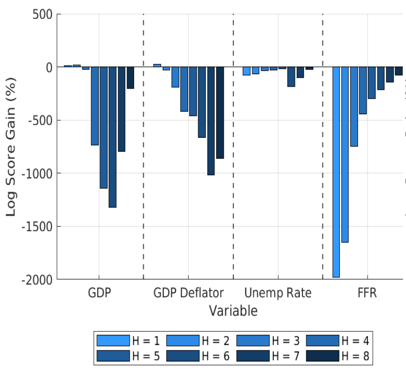


Large VAR

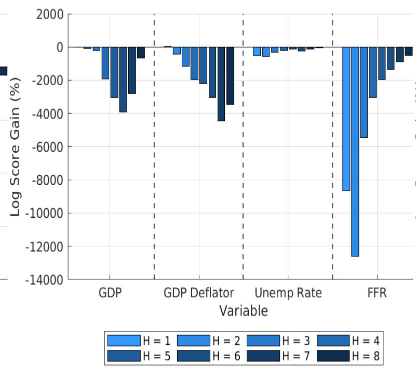


4. COVID

Small VAR



Medium VAR



Large VAR

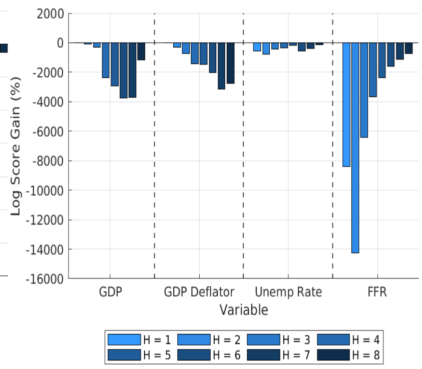


Table 9. Sample 1: p-values from AG test

Horizon	1	2	3	4	5	6	7	8
Small								
- GDP growth rate	0.0000	0.1332	0.5384	0.7248	0.5092	0.2620	0.0484	0.0021
- Inflation	0.0137	0.0342	0.0817	0.0111	0.0080	0.0212	0.0016	0.0008
- Unemployment rate	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000
- Federal funds rate	0.0006	0.0015	0.2864	0.6533	0.2165	0.0454	0.0031	0.0003
Medium								
- GDP growth rate	0.0001	0.2403	0.0998	0.0125	0.1556	0.0187	0.0349	0.0012
- Inflation	0.0042	0.0001	0.0000	0.0001	0.0000	0.0001	0.0000	0.0000
- Unemployment rate	0.0451	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000
- Federal funds rate	0.0553	0.0319	0.1402	0.2617	0.3685	0.4571	0.5402	0.6226
Large								
- GDP growth rate	0.3887	0.1736	0.1120	0.6937	0.0776	0.0494	0.0708	0.0041
- Inflation	0.0334	0.0005	0.0012	0.0000	0.0011	0.0014	0.0002	0.0000
- Unemployment rate	0.5429	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000
- Federal funds rate	0.2156	0.0225	0.1111	0.2883	0.4650	0.5750	0.6574	0.6916

A.2 A Forecast Accuracy Improvements with Alternative Forecast Aggregation Scheme

As an alternative to conditioning on a single REPS-based value of λ, I implement REPS as a posterior reweighting scheme. I retain the GLP posterior draws of λ and the associated predictive distributions, but replace the equal mixture weights used by GLP with REPS distortion weights. This preserves the posterior-mixture structure of GLP and assesses whether the REPS distortion has predictive content beyond the use of a single plug-in hyperparameter.

Let $\{\lambda_s\}_{s=1}^{S}$ denote the posterior draws of the tightness parameter obtained from the GLP. The standard GLP predictive density can be written as an equal-weight mixture.

$$p^{GLP}(y^{T+h}|Y) = \frac{1}{S}\sum_{s=1}^{S} p(y^{T+h}|Y, \lambda_s).$$

The REPS-reweighted predictive density keeps the same posterior draws but replaces the equal weights with REPS distortion weights:

$$p^{REPS-w}(y^{T+h}|Y) = \sum_{s=1}^{S} w_s^{REPS} p(y^{T+h}|Y, \lambda_s).$$

where

$$w_s^{REPS} = \frac{\exp\left[\eta \bar{L}_s(\lambda_s - \bar{\lambda}^{REPS})\right]}{\sum_{j=1}^{S}\exp(\eta \bar{L}_j(\lambda_j - \bar{\lambda}^{REPS}))}$$

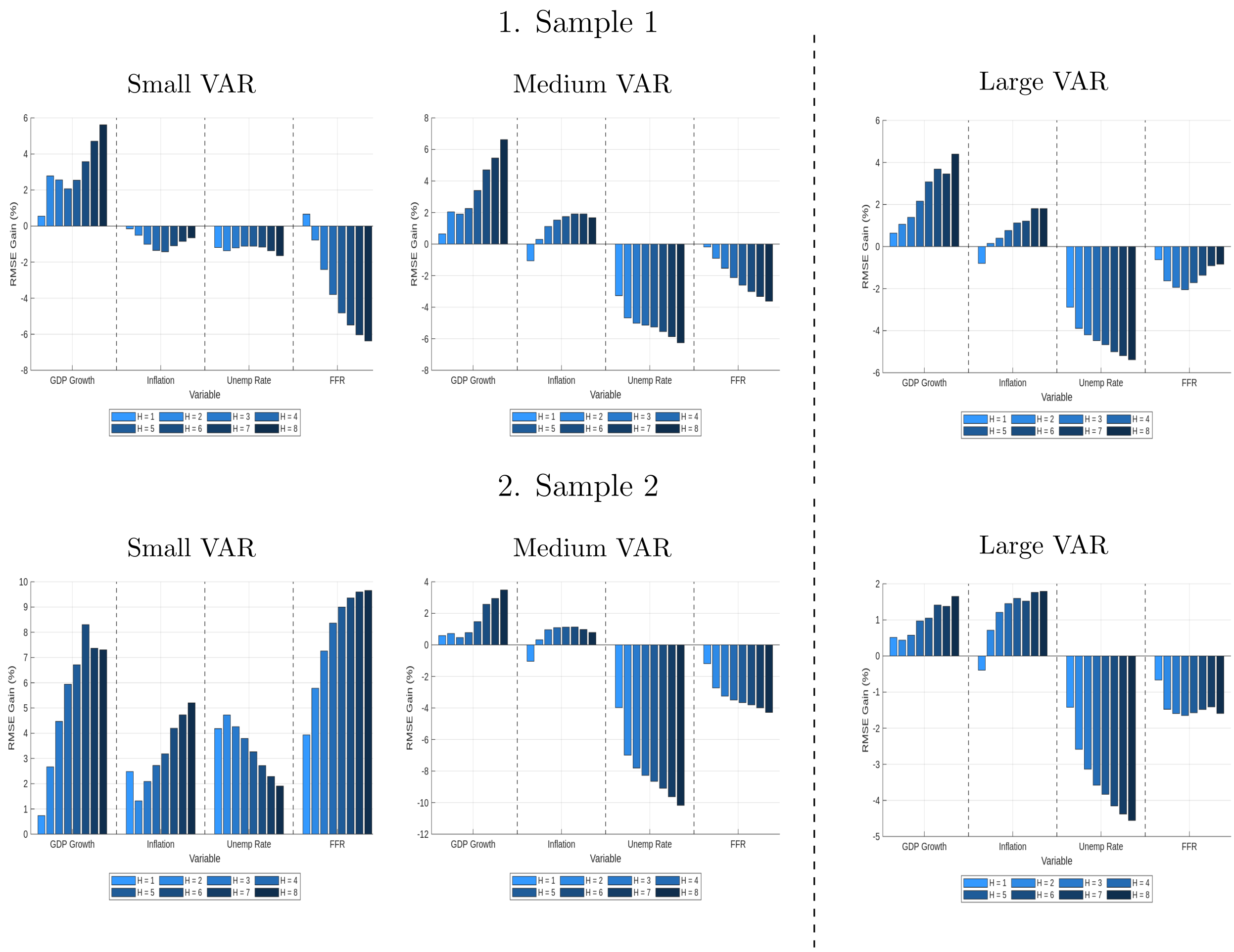
Here, $\bar{L}_s$ denotes the normalized marginal likelihood evaluated at λ_s , and η is the Lagrange multiplier associated with the entropy constraint. Since the REPS problem considered here minimizes the distorted posterior mean of λ, the multiplier is negative. As a result, posterior draws with relatively small values of λ_s receive larger weights when they are also supported by the marginal likelihood. The reweighted forecast therefore preserves the posterior-mixture structure of GLP while tilting posterior mass toward tighter priors.

Figures 9 and 10 report the forecasting performance of the posterior-reweighted REPS implementation. The point forecast results in Figure 9 are more modest than those from the baseline REPS plug-in implementation. RMSFE gains are concentrated in GDP growth and inflation, especially in small and medium VARs, while gains are weaker and often negative for the unemployment rate and the federal funds rate, particularly in medium and large VARs and at longer horizons.

The density forecast results in Figure 10 are more favorable. The REPS-reweighted forecasts deliver positive log score gains in many small and medium VAR specifications, especially for GDP growth and inflation. However, the improvements are not uniform. Density forecast losses remain more common for the unemployment rate, the federal funds rate, and during the COVID period, particularly in larger models.

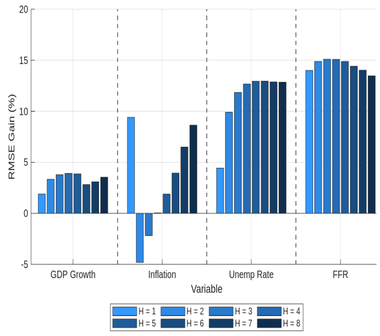
Overall, the reweighting exercise supports the interpretation that the REPS distortion contains predictive information beyond the plug-in implementation, but its gains are smaller and more variable-specific. Thus, the posterior-reweighted results are best viewed as a conservative robustness check rather than as the main forecasting specification.

Figure 9: Percentage Change in RMSFE

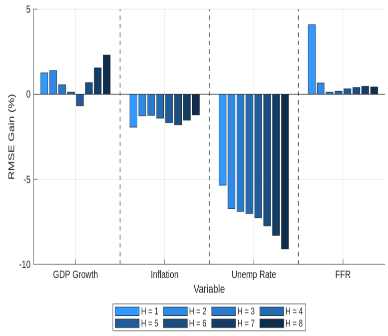


3. Financial Crisis

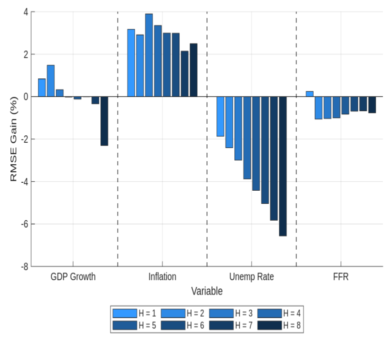
Small VAR



Medium VAR

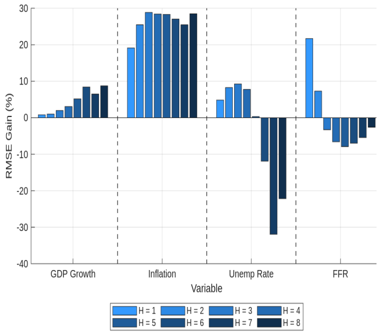


Large VAR

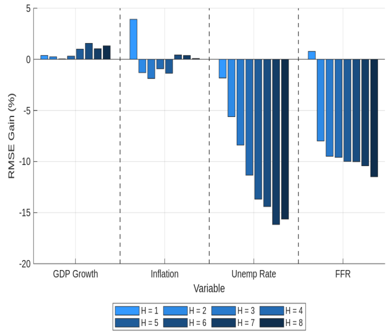


4. COVID

Small VAR



Medium VAR



Large VAR

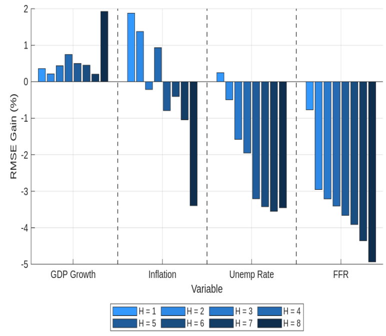
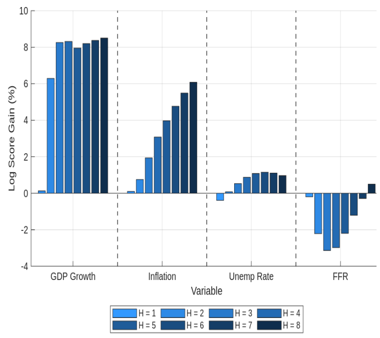


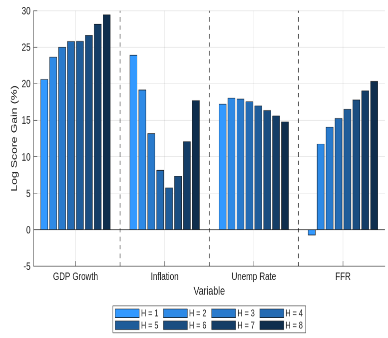
Figure 10: Log Socre Change in RMSFE

1. Sample 1

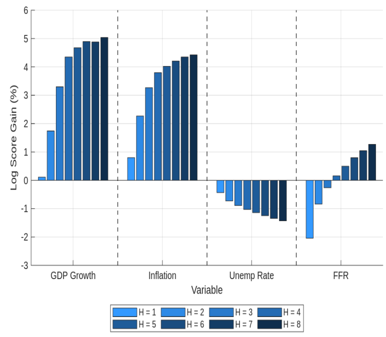
Small VAR



Medium VAR

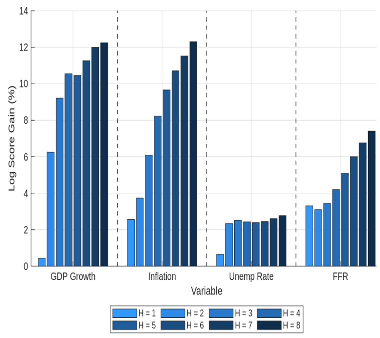


Large VAR

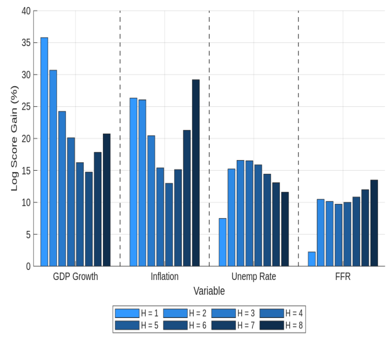


2. Sample 2

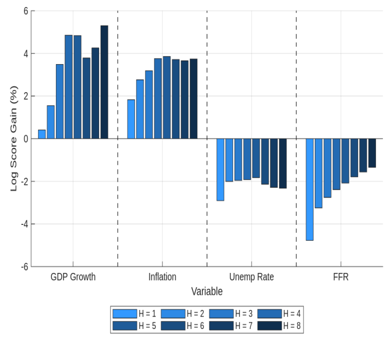
Small VAR

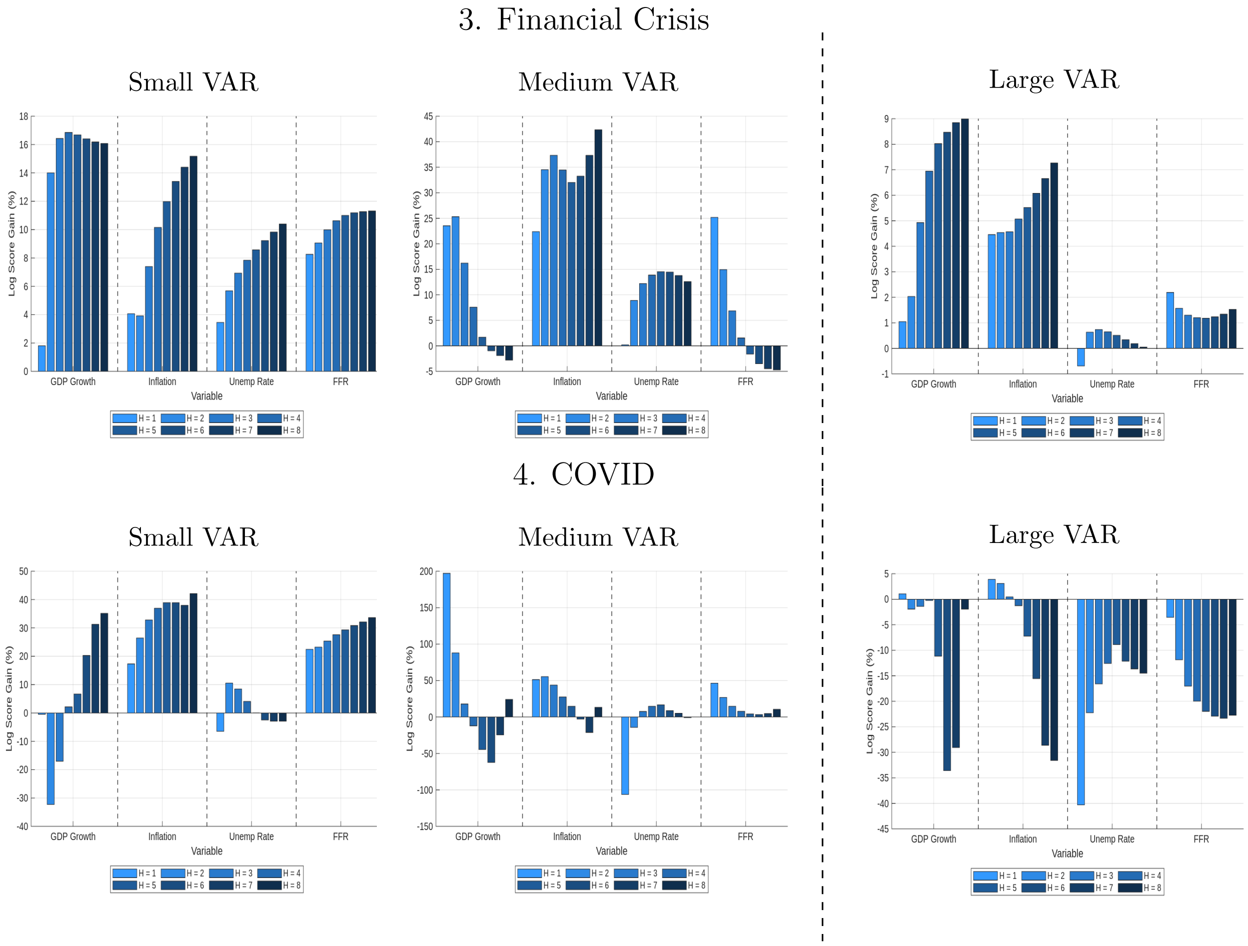


Medium VAR



Large VAR





B Forecast Accuracy Improvements with an Additional Prior

The *dummy-initial-observation prior* supplements the *Minnesota prior* by incorporating the belief that variables jointly tend to stay near their initial values. It achieves this by introducing dummy observations that anchor the system to its starting point.

$$y^{++} = \bar{y}'/\delta, \quad x^{++} = \left[\frac{1}{\delta}, y^{++}, \cdots, y^{++}\right]$$

The hyperparameter δ captures the tightness of the prior. As $\delta \to \infty$, the prior becomes uninformative. In the estimation, I treat the hyperparameter δ as an unknown parameter that needs to be estimated. Consequently, I impose a Gamma hyperprior with a mode of 1 and a standard deviation of 1, following the specification in Giannone et al. (2015).

After generating 100,000 posterior draws of λ and δ, I used their posterior means as the GLP hyperparameters and compared the resulting forecasts to those obtained using the corresponding REPS-based hyperparameters. As explained in Appendix A, forecast accuracy improvements from using REPS are expected to be smaller in this case, since the comparison no longer reflects hyperparameter variation.

Nonetheless, because REPS is allowed to optimize an additional hyperparameter, potentially improving out-of-sample forecast accuracy at the expense of in-sample fit, the forecast accuracy gains from REPS remain higher than when only λ is allowed to vary (Figure 9).

Likewise, the percentage change in log score from using REPS increases when both λ and δ are allowed to vary, compared to the case where only λ is varied (Figure 10).

Figure 11: Percentage Change in RMSFE

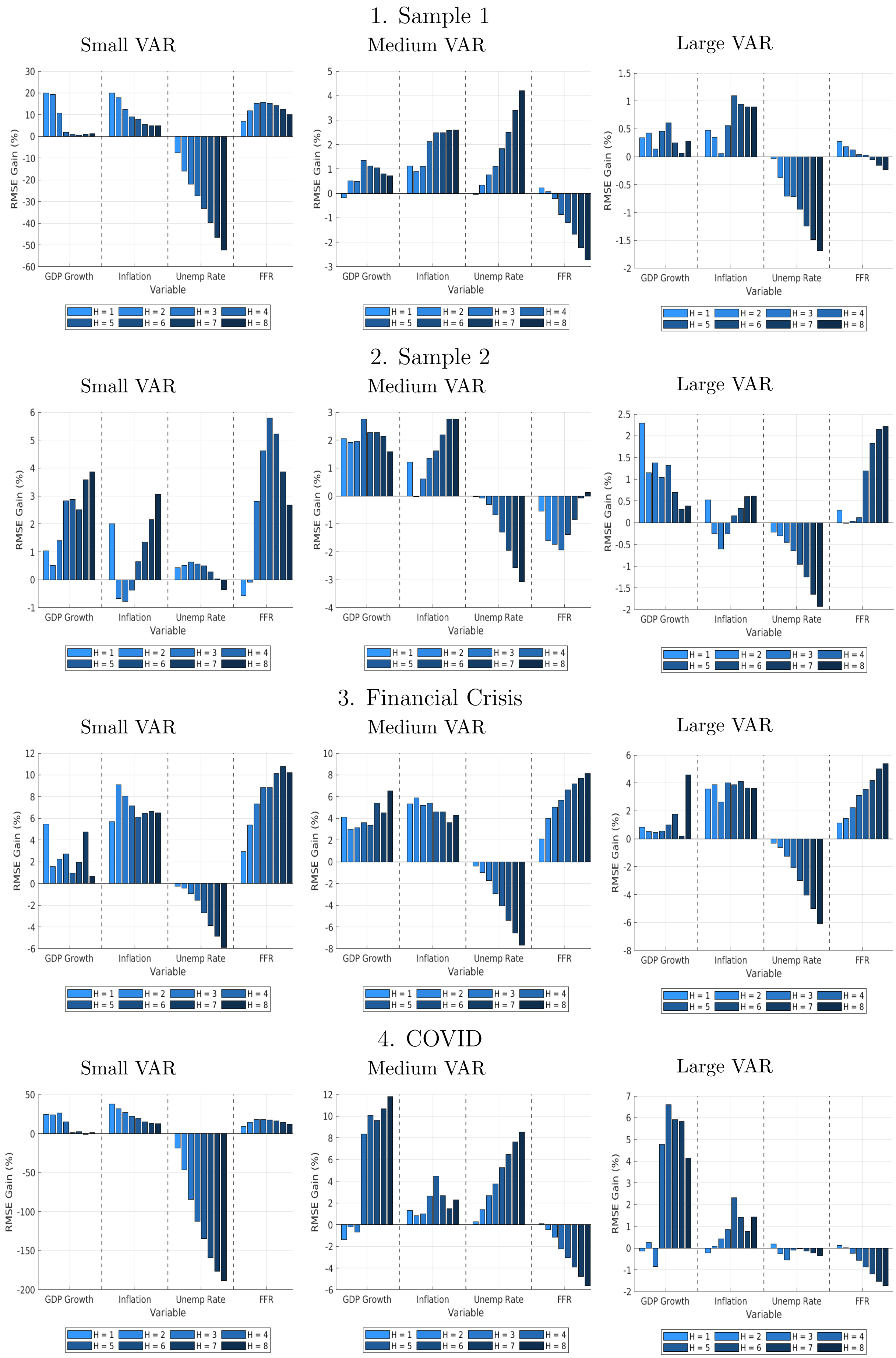
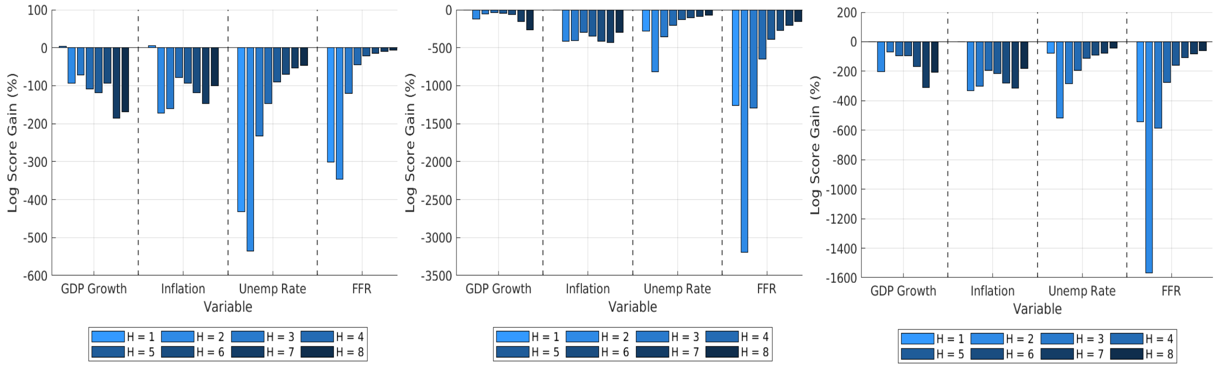


Figure 12: Percentage Change in Log Score

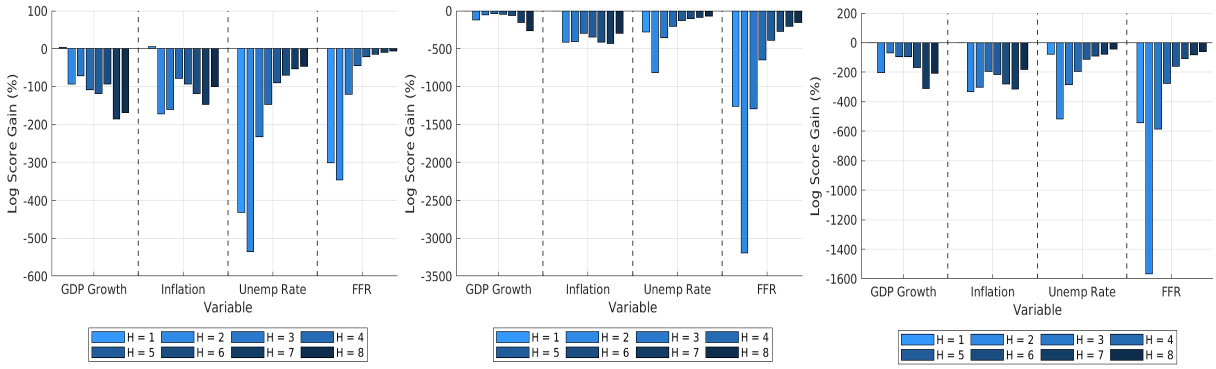
1. Sample 1

Small VAR | Medium VAR | Large VAR



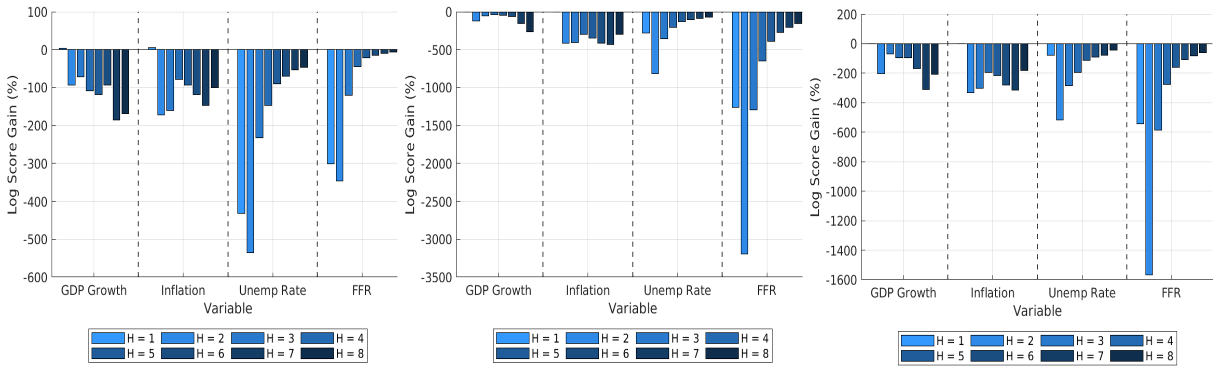
2. Sample 2

Small VAR | Medium VAR | Large VAR



3. Financial Crisis

Small VAR | Medium VAR | Large VAR



4. COVID

Small VAR | Medium VAR | Large VAR